

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with X-Regularization

Qi Feng¹, Rongjie Lai², Di Qi², and Xuda Ye²

¹Department of Mathematics, Florida State University, Tallahassee, FL 32306

²Department of Mathematics, Purdue University, West Lafayette, IN 47907

June 26, 2026

Abstract

Forward KL training of a normalizing flow Boltzmann generator is mass-covering and usually trains well, but two weaknesses limit it, both rooted in the annealing chain that supplies its target samples and is initialized at the flow’s own pushforward. First, the chain mixes only among the modes the flow already covers, so the objective can silently miss the rest and report a fake effective sample size, where the importance weights on the reached support look benign while target mass is absent. Second, when the chain returns noisy or biased samples the stability and accuracy of training suffer, and the objective constrains only the mean of the log-ratio under the target, leaving its spread free to vary. A third, smaller mechanism rides alongside: where the target vanishes the gradient carries no signal about the mass the flow leaks there. We address all three with a family of X functionals that penalize the pairwise difference of the flow’s own log-ratio under a chosen non-negative weight, a localized constraint that cancels the unknown normalizer and vanishes when the log-ratio is constant on the support of that weight. Three weights take complementary roles: one for the shape of the log-ratio under the target, one for mode discovery against a wide-coverage measure, the quench and temper measure built once before training, and one for the leakage outside the support of the target. A balanced loss combines the three, folding mode discovery and leakage suppression into a single pairwise integral alongside the forward KL and the shape term; it augments rather than replaces the forward KL, and the shape term reuses the forward KL’s forward pass while the mixture adds a further sampling batch. A Fisher–Rao gradient-flow view shows the forward KL still decays along the shape-regularized flow and, under a bound on the regularization strength, the accuracy floor left by biased samples can be lowered. We build on the loss an adaptive-temperature Boltzmann generator and study it on model problems and molecular potential energies. Heavy-tailed targets remain hard, since the quench tends to miss their thin outer mass, and on a singular interaction core, where the difficulty is not multimodality, the gains shrink.

1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$, known only up to its normalization constant, is a recurring problem in molecular simulation, Bayesian inference, and statistical physics. When μ is high-dimensional and multimodal, its mass splits among basins of U separated by barriers, and a local Markov chain Monte Carlo (MCMC) sampler that proposes small moves is trapped: the time to cross a barrier of free-energy height ΔF grows like $e^{\Delta F}$, so within any practical budget the chain explores the basin it began in and leaves the rest of the mass unseen (Metropolis et al., 1953; Hastings, 1970). The estimates such a chain returns are biased toward that basin, and the bias is silent, since the chain reports nothing about the mass it never visited.

The classical responses soften the barriers rather than remove them. Parallel tempering (PT) and replica exchange run replicas at a ladder of temperatures and swap configurations between neighbors, carrying a basin crossed hot down to the temperature of interest (Swendsen and Wang, 1986; Geyer, 1991; Hukushima and Nemoto, 1996). Annealed importance sampling (AIS) bridges a tractable proposal to μ along intermediate distributions while accumulating importance weights (Neal, 2001), sequential Monte Carlo (SMC)

carries a weighted particle population along the same kind of path (Doucet et al., 2001; Del Moral et al., 2006), and nested sampling recasts the normalizing integral as a one-dimensional quadrature over a rising likelihood threshold (Skilling, 2006). A line meeting the learned-sampler program partway replaces the fixed hot reference by a map, sharpening the swap schedule, tuning the endpoint, or training a flow to carry the hot reference toward the cold target (Syed et al., 2022; Surjanovic et al., 2022; Zhang et al., 2025); this line is the closest to ours; we differ in that our mode discovery is built off the ladder, rather than carried by the tempering chain itself. These methods are the workhorses of multimodal sampling and remain the references against which a learned sampler is judged. They share one limit: they move mass only between basins a chain can already traverse within budget, so a basin no replica reaches contributes nothing, and raising the temperature enough to bridge every barrier flattens the structure one is trying to resolve (Woodard et al., 2009; Chehab et al., 2025). We do not discard this machinery; we use it. Our training samples come from an AIS ladder, and the wide-coverage measure that drives mode discovery is built by a quench and temper construction that borrows the temperature-annealing idea while avoiding the reliance on a chain crossing every barrier; we supply the modes the chain cannot reach from a separate source.

Against this classical backdrop, Boltzmann generators (BG) replace the chain by a learned transport map (Noé et al., 2019). A normalizing flow is an invertible parametric map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018), trained so that the pushforward $\nu := (G^{-1})_{\#}\mu_0$ of an easily samplable reference $\mu_0 \propto \exp(-U_0)$, typically a Gaussian, approximates μ ; by convention G maps target to source, $G_{\#}\mu \approx \mu_0$, so G^{-1} draws a batch of independent samples by pushing μ_0 through it in one forward pass. Invertibility gives ν in closed form by the change of variables, so each sample carries its own importance weight μ/ν against the unnormalized target. Reweighting then turns the samples into Monte Carlo estimates with controlled finite-sample error, and the effective sample size (ESS) of those weights is a standard diagnostic of how well the flow matches μ on the support it has reached (Liu, 2001; Doucet et al., 2001; Elvira et al., 2022).

The choice of training objective then divides the field. The original BG (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse Kullback–Leibler (KL) divergence

$$(\text{reverse}) \quad \text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (1)$$

which needs only samples from ν and energy evaluations, and is therefore cheap. It is also mode-seeking (Wainwright and Jordan, 2008): on multimodal μ the flow tends to collapse onto a subset of the modes and cannot recover from a mode it never visits, since a missed mode contributes no gradient (Wu et al., 2020; Felardos et al., 2023). Several lines soften this without removing it, interleaving MCMC kernels with the deterministic layers, folding the symmetries of μ into the architecture, lowering the gradient variance near the optimum, or annealing from a hot landscape toward μ (Wu et al., 2020; Köhler et al., 2020; Vaitl et al., 2022; Schopmans and Friederich, 2025); none restores a dropped mode, because the reverse KL places no sample where ν has no mass. Our $X_{\bar{\nu}}$ term below plays the leakage-suppression role one wants from a reverse KL-like penalty, restraining the mass ν places where μ has none, but reads it from a stop-gradient copy of ν , so the flow’s own sampling is never differentiated through, the source of the reverse KL’s instability on sharp targets.

The opposing objective minimizes the forward KL,

$$(\text{forward}) \quad \text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is mass-covering, diverging wherever ν misses mass that μ carries (Wainwright and Jordan, 2008), but needs samples from μ , the very object one is trying to produce. Existing work breaks this circularity by choosing where the target samples come from, and each choice fixes what the method can see: maximum likelihood on configurations from a reference thermodynamic state (Wirnsberger et al., 2020), a flow trained on a running chain and used as a global proposal (Gabrié et al., 2022), fitting to a molecular dynamics (MD) trajectory (Klein et al., 2023), amortizing coverage across many molecules through a shared architecture (Klein and Noé, 2024), or supplying the intractable gradient from a persistent chain or an ℓ_2 fit to coupled targets (Naesseth et al., 2020; Rehman et al., 2026). In every case the forward KL is silent about mass its sample source never reaches: a mode absent from the reference data, the running chain, or the trajectory set

leaves no trace in the gradient, exactly as a missed mode is invisible to a chain. These remedies change the training signal or amortize coverage across systems; by contrast we keep the bare forward KL and confront the circularity directly, adding the X functional below, which reuses the same samples on a single target. Our AIS ladder supplies the target samples, and because that ladder inherits the classical limit, the regularizer draws on an independent sample source built to cover modes the chain misses.

The line closest to ours couples an annealing ladder to flow training so the flow improves the sampler that trains it. The flow annealed importance sampling bootstrap (FAB) (Midgley et al.) replaces the forward KL by the α -divergence at $\alpha = 2$, whose integrand is proportional to μ^2/ν and whose minimizer also minimizes the variance of the importance weights; it draws training points from AIS chains targeting this minimum-variance surrogate and recycles them through a prioritized replay buffer. Related work weaves a learned flow into each rung of an SMC annealing path, reweights down a temperature ladder, adds an inference-time SMC correction at molecular scale, or curbs the mass teleportation these geometric paths allow (Arbel et al., 2021; Matthews et al., 2022; Schopmans and Friederich, 2025; Tan et al., 2025; von Klitzing et al., 2026). We take this AIS-coupled forward KL line as our point of departure and depart from it on three counts, set side by side in Remark 3. First, we keep the bare forward KL and add the X functional rather than change the divergence, so the minimizer is unchanged and the regularizer reuses the same samples and the same autograd graph. Second, our AIS ladder targets μ itself and is score-free in the flow: every chain moves under the target score $-\nabla U$, the flow enters only through evaluations of the log-ratio, and no replay buffer is kept, whereas the surrogate μ^2/ν has a score that carries the positional gradient of the pushforward density at every chain step. Third, mode discovery is delegated to a quench and temper measure $\hat{\mu}$ rather than asked of a chain that would still have to cross the barriers of μ . The trade-offs are real: the order-2 Rényi divergence behind μ^2/ν carries a stronger tail-coverage incentive than the forward KL, and with a sufficient annealing budget FAB needs no separate wide-coverage construction; and where an inference-time SMC pass buys scale, we keep the correction inside the tractable per-sample importance weight, so the score-free ladder and the absent replay buffer leave a small memory footprint and let the method run on a single consumer GPU. The off-policy log-dispersion regularizer penalizes a scalar, variance-type spread of $\log(\mu/\nu)$ (Schopmans et al., 2026); the X functional below instead penalizes the localized pairwise variation of $\log(\mu/\nu)$ under a chosen weight, and it is this weight that lets one family take the three complementary roles of mode discovery through $\hat{\mu}$, leakage suppression through $\bar{\nu}$, and shape control through μ from a single pairwise integral.

A separate tradition drops the bijection and trains a stochastic process directly: a controlled stochastic differential equation, a denoising diffusion, or a flow-matching velocity carries the reference to μ along a continuous-time path, learning the drift rather than an invertible map (Zhang and Chen, 2022; Vargas et al., 2023; Berner et al., 2024; Nüsken and Richter, 2021; Richter and Berner, 2024). Energy-only variants such as iterated denoising energy matching (iDEM) alternate a sampling loop with a score- or velocity-fitting loop and a replay buffer (Akhound-Sadeh et al., 2024; Woo and Ahn, 2024), the non-equilibrium transport sampler augments AIS with a learned drift and so sits between the AIS-coupled and diffusion lines while still offering no tractable terminal density (Albergo and Vanden-Eijnden, 2025), and successors scale the idea to larger many-body systems (Havens et al., 2025; Akhound-Sadeh et al., 2025). These samplers are expressive and train from the energy alone. The cost is decisive for our purpose: once the bijection is gone, the terminal density ν is no longer available in closed form, so the importance weight μ/ν , the ESS read from it, and the reweighting that corrects residual bias are out of reach without a further per-sample ordinary-differential-equation solve (He et al., 2025). We commit to invertible flows for that reason: the invertibility of G keeps the log-ratio $\log(\mu/\nu)$ exact and inexpensive per sample, and with it the finite-sample importance-sampling pipeline on which both the forward KL and the X functional rest. The two lines are complementary rather than competing, one buying scale through dynamics at the cost of a tractable density, the other keeping the density and the reweighting at the cost of invertibility. Table 1 places these training lines side by side.

What is new. The forward KL inherits the classical limit through its sampler: AIS initialized at ν mixes only among the modes ν already covers. This leaves the objective with two core weaknesses, which organize the rest of the paper, together with a third, smaller mechanism the same regularizer happens to address. The first weakness is silent missed modes, the fake ESS pitfall. The ESS is a faithful diagnostic only when μ and ν are weighed on the same support; if ν misses a mode of μ , the chain initialized at ν rarely escapes the basins ν already covers, the weights on the reached support can be near-uniform, and the ESS can sit

	objective	AIS required	importance weights	mode source	replay buffer	score-free flow
reverse KL	$\text{KL}(\nu\ \mu)$	–	✓	none	–	✓
forward KL	$\text{KL}(\mu\ \nu)$	✓	✓	AIS chain	–	✓
FAB	$\alpha=2$ divergence, μ^2/ν	✓	✓	AIS chain	✓	–
iDEM	energy matching	–	–	sampling loop	✓	–
ours	$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$	✓	✓	$\hat{\mu}$, no chain	–	✓

Table 1: Where the X-regularized forward KL flow sits among Boltzmann generator training lines. Entries are qualitative; ✓ marks a property the line has by construction, and – its absence.

close to its ceiling while a part of the mass of μ is absent (Noé et al., 2019; Midgley et al.). The second weakness is accuracy under imperfect target samples: the AIS samples are noisy or biased, most of all at small batch and on early bridges, and because the objective constrains only the mean of $\log(\mu/\nu)$ under the target, leaving its spread free to vary, it inherits that sample bias with no mechanism to trim it. The smaller mechanism is leakage: where μ vanishes, the forward KL integrand $\mu \log(\mu/\nu)$ vanishes too, so the gradient carries no signal about mass that ν places outside the support of μ . The missed-mode weakness in particular is not cured by sampling the modes ν already reaches more accurately.

Our remedy is to regularize the forward KL with a family of X functionals X_ω , named for the pairwise variation of the log-ratio $z = \log(\mu/\nu)$ they penalize. For a non-negative weight ω on $\mathbb{R}^d$, not necessarily normalized, let

$$\text{X}_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

against which both arguments are integrated. The idea is direct: matching ν to μ is the same as flattening z , and the pairwise difference cancels the unknown normalizer of μ . Where the forward KL constrains $\nu \approx \mu$ only in a global, Jensen-averaged sense, X_ω penalizes the variation of z wherever ω places mass and localizes the constraint to the support of ω . It vanishes whenever z is constant on that support, so its minimizer is insensitive to the precise shape of ω , and for $\omega = \mu$ its integrand is built from the same log-ratio field as the forward KL, reusing the same samples and the same autograd graph.

Three choices of ω , with complementary roles, jointly address these failures. With $\omega = \mu$, the functional X_μ controls the spread of z under the target where the forward KL controls only its mean, tightening the density match once ν has reached the modes and trimming the bias inherited from imperfect annealed samples. With $\omega = \hat{\mu}$, a coarse, deliberately over-spread approximation of μ , the functional $\text{X}_{\hat{\mu}}$ drives mode discovery: by the insensitivity above, the only role of $\hat{\mu}$ is to place mass on modes and tails the current ν has not yet covered, and the gradient then pulls the flow toward them. We build $\hat{\mu}$ once, before training, with a quench and temper construction that bridges basins, collapses onto mode centers, and rejuvenates with a short Langevin chain; it errs toward broad coverage, and inherits a blind spot on heavy outer tails. With $\omega = \bar{\nu}$, a stop-gradient copy of the current pushforward, the functional $\text{X}_{\bar{\nu}}$ plays the reverse KL role without its instability, penalizing the mass ν places outside the support of μ and so suppressing the leakage the other two terms cannot see. Our preferred loss combines all three with the forward KL,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda \text{X}_\mu(\mu\|\nu) + \text{X}_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu), \quad (4)$$

with weights $\lambda, \alpha, \beta \geq 0$ of order one, the third term being X_ω at the mixture weight $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$. Putting the mixture inside a single pairwise integral does more than add the two self-terms: expanding the squared base measure also generates a cross term that anchors the in-support and out-of-support log-ratios to each other. Written through the change of variables, every term reads off the single scalar field along the flow,

$$z(y) = U_0(G(y)) - U(y) - \log |\det J_G(y)|, \quad (5)$$

up to an additive constant that cancels in each pairwise difference, so the loss in terms of the flow map G is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu} [z(y) - z(y')]}_{\text{X}_\mu(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}} [z(y) - z(y')]}_{\text{X}_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (6)$$

The whole loss is computed from one forward pass of G on a batch: the X_μ term reuses the forward pass and batch of the forward KL, while the mixture term draws a further sampling batch. None of the X terms moves the minimizer, since each vanishes at $\nu = \mu$.

Contributions. This paper makes the following contributions. We introduce the X functionals X_ω , which augment the forward KL by penalizing the pairwise variation of the flow’s own log-ratio $z = \log(\mu/\nu)$, and we study three instances of complementary role: X_μ , $X_{\hat{\mu}}$, and $X_{\bar{\nu}}$, together with the balanced loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, which folds mode discovery and leakage suppression into one pairwise integral and serves as a default; re-tuning the weights away from it did not improve our results. The construction is also lightweight in practice, on three counts collected here: the loss reads off a single forward pass of the flow, no replay buffer is kept, and the AIS ladder is score-free in the flow; the mixture adds a further sampling batch beyond a plain forward KL step. The payoff is a modest memory footprint that lets the method run on a single consumer GPU, with no replay buffer and a score-free ladder, unlike the buffer-based and score-coupled designs. We give a Fisher–Rao analysis showing that the forward KL still decays along the X_μ -regularized flow under no structural assumption on μ , and that, under a bound on the regularization strength, the X_μ term can lower the accuracy floor that biased AIS samples leave behind, the regime where the bare forward KL is most fragile and the training batch is small. We pair the ESS with a coverage diagnostic (Naeem et al., 2020) that measures how much of the wide-coverage measure the trained ν reaches, so that the two together check density calibration on the reached support and that no mode the wide-coverage measure reaches is silently missed, though the diagnostic inherits the quench’s blind spot for thin heavy tails. Finally, we build on the loss an adaptive-temperature Boltzmann generator that grows its temperature ladder from the per-stage effective sample size of a classical SMC run; for molecular potentials, a sharpening reweight pulls each stage toward the physical potential, and we study the method across smooth multimodal targets, a high-dimensional multi-well family, and lattice field theories before applying it to molecular potential energies; we are explicit about its limits: the quench tends to miss the thin outer mass of heavy-tailed targets, and on a singular interaction core the difficulty is not multimodality.

Organization. Section 2 studies the Fisher–Rao gradient flow of the regularized loss, its convergence, and its accuracy floor under biased target samples. Section 3 turns the loss into a training algorithm for the flow map G , detailing the AIS that targets μ , the quench and temper construction of the wide-coverage measure $\hat{\mu}$, and the ESS and coverage diagnostics. Section 4 builds the adaptive-temperature Boltzmann generator on top of this loss. Section 5 reports the numerical study on model problems, and Section 6 applies the method to molecular potential energies. Section 7 concludes.

2 Fisher–Rao gradient flow with X functionals

To gauge how the regularizer shapes training we view ν as a density evolving along the Fisher–Rao (FR) gradient flow of the loss $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$, the continuous-time idealization whose convergence rate is a proxy for trainability. Write $w = \mu/\nu$ for the importance weight and $z = \log w$ for the log-ratio. In the FR geometry, unlike the Wasserstein-2 one, the forward KL admits an unconditional exponential decay, and the regularizer adds a nonlocal birth–death modulation driven by where w ranks against the population of weights under μ . Differentiating $\text{KL}(\mu\|\nu_t)$ along the flow and reducing the X_μ term by a rank-correlation identity gives the dissipation

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = -\chi^2(\mu\|\nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right], \quad (7)$$

with χ^2 the chi-squared divergence; both terms are non-negative, so the regularizer only speeds the decay. Since $\chi^2(\mu\|\nu_t) \geq \text{KL}(\mu\|\nu_t)$, Grönwall’s lemma yields the exponential rate

$$\text{KL}(\mu\|\nu_t) \leq e^{-t} \text{KL}(\mu\|\nu_0), \quad t \geq 0, \quad (8)$$

under no structural assumption on μ — no log-concavity, spectral gap, or growth condition — in contrast to the Wasserstein-2 geometry, where the forward KL is not displacement convex and no such rate is available.

The rate is not a claim that forward KL is intrinsically faster: it is a population statement on the space of densities, agnostic to the cost of producing the target samples that AIS must supply, and the contribution of X_μ to (7) is structural — it amplifies mass where ν_t undercovers μ — rather than a faster transient.

The convergence above assumes the outer expectation is taken under the exact target. In practice AIS supplies only an approximation $\tilde{\mu}$, and the realized loss replaces μ by $\tilde{\mu}$ in its outer integrands. At $\lambda = 0$ a stationary point is $\nu^* = \tilde{\mu}$: pure forward KL inherits the AIS bias verbatim, leaving the residual error $\text{KL}(\mu\|\tilde{\mu})$ as a hard floor. The shape term lowers that floor.

Theorem 1 *Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, \frac{1}{2}]$. Then every stationary density ν^* of the X -regularized loss $\tilde{\mathcal{L}} = \text{KL}(\mu\|\cdot) + \lambda X_\mu(\mu\|\cdot)$ taken under $\tilde{\mu}$ satisfies*

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x)\tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (9)$$

The rank-based feedback inflates ν^* where it undercovers μ and deflates it elsewhere, pulling the stationary density off $\tilde{\mu}$ back toward μ by an amount set by the spread of the importance weights μ/ν^* across the support of $\tilde{\mu}$. So X_μ is not merely a stabilizer for the forward KL flow: it lowers the asymptotic accuracy floor left by biased AIS samples — the practically relevant regime, with small batches and early bridges — and the improvement persists well beyond the worst-case threshold $\lambda \leq \frac{1}{2}$. The full FR-gradient derivation, the rank-correlation lemma, the convergence proof, and the accuracy proof are given in Supplement S1.

3 Flow training with X functionals

This section turns the abstract loss of the introduction into a concrete training algorithm for the flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is the easily samplable reference and $\mu \propto \exp(-U)$ is the target. We parameterize $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ so that $G_{\#}\mu \approx \mu_0$ — mapping target to source, so that no inversion is required during training — and write $\nu := (G^{-1})_{\#}\mu_0$ for the induced approximation to μ . We first assemble the training loss and read off the two sampling ingredients it requires (Section 3.1), then supply them in turn: AIS of μ (Section 3.2), the quench and temper construction of the wide-coverage measure $\hat{\mu}$ (Section 3.3), and finally the ESS and coverage diagnostics used to evaluate the trained flow (Section 3.4).

3.1 The X-regularized training loss

By the change-of-variables formula the pullback density of μ_0 under G is $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, so the log-ratio between the target and the model along the flow is the single scalar field $z(y)$ of (5). Both ingredients of the loss are elementary functionals of z . The forward KL is its mean under the target, $\text{KL}(\mu\|\nu) = \mathbb{E}_{y \sim \mu}[z(y)] + \text{const}$ (cf. (2)), and each X functional X_ω (3) is the mean absolute pairwise variation of z under a non-negative weight ω , $X_\omega(\mu\|\nu) = \mathbb{E}_{y, y' \sim \omega}[|z(y) - z(y')|]$ when ω is normalized (in general it is the double integral (3)). Combining the shape term $\omega = \mu$ with the balanced mixture $\omega = \alpha\hat{\mu} + \beta\bar{\nu}$ that handles mode discovery and leakage suppression, the total training loss in terms of the flow map G is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_\mu(\mu\|\nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha\hat{\mu} + \beta\bar{\nu}}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (10)$$

We call the objective (10), previewed as (6) in the introduction, the *X-regularized forward KL* (the X-regularization of the title). We stress the two pushforward measures appearing in (10). The model distribution $\nu = (G^{-1})_{\#}\mu_0$, with density $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, is the flow’s live approximation to the target μ and depends on the trainable parameters of G . The barred measure $\bar{\nu}$ is the *same* pushforward frozen at the current parameters, a stop-gradient copy of ν , which enters the mixture term only as a sampling weight and is never differentiated through.

The three sampling measures entering (10) address complementary failure modes of the bare forward KL:

- μ : *accuracy at small batch*. The shape term X_μ compares the importance weights pairwise across the batch and lowers the asymptotic accuracy floor under biased AIS samples (Section 2); the gain is largest when the batch size is small and the resulting KL gradient is noisy.

- $\hat{\mu}$: *mode discovery*. The wide-coverage measure built by quench and temper (Section 3.3) places sample pairs across modes that AIS from the current pushforward cannot reach, pulling ν toward mass the bare loss never sees.
- $\bar{\nu}$: *leakage suppression*. The frozen pushforward penalizes the mass the flow places outside the support of μ , preventing the overfit that target samples alone cannot detect: where μ provides no samples, the bare forward KL provides no signal.

None of the three changes the minimizer, each X term vanishing at $\nu = \mu$; what they add to the existing forward KL is numerical robustness, not a new objective.

Remark (Robustness, not speed) *As discussed in Section 2, the exponential rate (8) is a statement about the idealized FR flow: the discrete Adam dynamics carries no such guarantee and may take many steps to converge. Moreover, at large batch the margin over the bare forward KL can be minor, since a large batch is already a powerful safeguard against mode collapse; the gap is widest in the small-batch regime (Table 6).*

All three terms are computed in a single forward pass: evaluate $\mathbf{z} = (z(y_1), \dots, z(y_B))$ once on a batch, take $\text{mean}(\mathbf{z})$ for the forward KL, and $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ for a random permutation σ of the batch for each X term. Backpropagation passes through $\mathbf{z}$ exactly once, identical to a vanilla forward KL step, so the X functionals add negligible extra cost.

The loss draws on three measures, of which only two must be built. The $\bar{\nu}$ batch is free: it is the raw pushforward $\bar{y}_i = G^{-1}(x_i)$ of source samples $x_i \sim \mu_0$, taken with the gradient through G detached (a stop-gradient); this detached sampler keeps the loss numerically stable on sharp targets, where differentiating through the sampling chain, as a true reverse KL gradient does, would explode. The other two require construction: samples of the target μ for the forward KL and X_μ terms, supplied by AIS (Section 3.2), and samples of the wide-coverage measure $\hat{\mu}$ for the mixture term, supplied by quench and temper (Section 3.3). Therefore, if the same batch size is used for approximating the measures μ and $\alpha\hat{\mu} + \beta\bar{\nu}$, the raw training cost of the X-regularized loss is approximately twice that of the forward KL. Inference uses the same flow architecture, and is hence identical for both methods.

Although (λ, α, β) are three free parameters in principle, the X functional is by construction on the same scale as the forward KL, so any choice of order 1 works. Two settings are worth highlighting. The *X_μ -only hyperparameters* $\lambda = 1, \alpha = \beta = 0$ add only X_μ on top of the forward KL; this term reuses the AIS samples and autograd graph of the forward KL, costing only an extra permutation and element-wise difference per batch, and it suffices whenever the source μ_0 already overlaps every mode of μ . The *balanced hyperparameters* $\lambda = 1, \alpha = \beta = \frac{1}{2}$ turn on the mixture term, an equal-weight draw from $\hat{\mu}$ and $\bar{\nu}$: the $\hat{\mu}$ half pulls ν toward modes discovered by quench and temper that AIS alone cannot reach, while the $\bar{\nu}$ half restrains the mass the flow places outside the support of μ . We recommend the balanced hyperparameters as the default whenever μ_0 may not cover all modes of μ , and use them throughout the experiments; re-tuning (λ, α, β) away from these defaults did not improve results on any of our benchmarks.

3.2 Sampling μ by annealed importance sampling

Evaluating the μ -expectations in (10) requires samples from μ , which are not directly available. The natural surrogate draws $y_1, \dots, y_B$ from the pushforward $\nu = (G^{-1})_{\#}\mu_0$ and reweights by

$$w(y) = \frac{\mu(y)}{\nu(y)} = \exp(z(y)) = \exp\left(U_0(G(y)) - U(y) - \log |\det J_G(y)|\right), \quad y \in \mathbb{R}^d, \quad (11)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ and ν typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of low quality. AIS bridges this gap by inserting a ladder of $M + 1$ distributions

$$\pi_m(y) \propto \nu(y)^{1-\frac{m}{M}} \mu(y)^{\frac{m}{M}} = ((G^{-1})_{\#}\mu_0)(y) w(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (12)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each stage of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling.

Algorithm 1: AIS from $(G^{-1})_{\#}\mu_0$ to μ

Input: samples $y_1, \dots, y_B$ from $(G^{-1})_{\#}\mu_0$, ladder length M , intermediate distributions π_m (12)
Output: samples $y_1, \dots, y_B$ approximately distributed as μ

- 1 **for** $m = 1, \dots, M$ **do**
 - // reweight along the geometric path
 - 2 Compute the incremental weights on $y_1, \dots, y_B$, $w(y_i)^{\frac{1}{M}} = \pi_m(y_i)/\pi_{m-1}(y_i) = \exp(z(y_i)/M)$;
// resample
 - 3 Resample from the weighted set, $y'_1, \dots, y'_B \leftarrow \text{resample}(y_{1:B}, (w(y_i)^{\frac{1}{M}})_{i=1}^B)$;
// rejuvenate under the target
 - 4 Diffuse the resampled duplicates by a short Langevin chain on μ , $y_{1:B} \leftarrow \text{langevin}(y'_{1:B}, U)$;
- 5 **end**

In particular, when G is the identity map the proposal collapses to the source, $(G^{-1})_{\#}\mu_0 = \mu_0$, the Jacobian term drops out of w , and Algorithm 1 reduces to classical SMC bridging μ_0 to μ .

In the general case, the Langevin step nominally targets the intermediate distribution π_m , and so requires its score $\nabla_y \log \pi_m(y)$. Through the pushforward density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$, this score carries the term $\nabla_y \log |\det J_G(y)|$, whose evaluation needs the Jacobian $J_G(y)$ itself (not merely its determinant, which the flow already provides), and is therefore expensive for a deep map. In practice we instead diffuse the resampled particles directly under the target μ , whose score $\nabla_y \log \mu(y) = -\nabla U(y)$ is cheap. Since the per-stage reweighting already carries the exact π_m correction along the geometric path (12) and μ is the ultimate target, the only effect of the substitution is on intermediate mixing, not on the weighted target: the Langevin kernel serves only to decorrelate the duplicates left by resampling.

Remark (Choosing the ladder length M) For simple targets close to the source, $M = 1$ suffices. The choice $M = d$ is typical but not necessary in every test — an empirical rule by which the ladder length scales with the dimension of the deformation the flow must bridge, of which d is the worst case. For instance, annealing $\mu_0 \propto e^{-|x|^2}$ to $\mu \propto e^{-2|x|^2}$ on $\mathbb{R}^d$ deforms all d coordinates at once, and a ladder of $M = d$ steps keeps the weighted empirical distribution at a finite ESS that does not degrade with d ; when the deformation concentrates on fewer directions, a much shorter ladder suffices.

Remark (Comparison with FAB) FAB (Midgley et al.) also couples AIS to flow training, but places the flow inside the chains. Its transitions target the mode-seeking surrogate μ^2/ν , whose score

$$\nabla_y \log \frac{\mu^2(y)}{\nu(y)} = -2\nabla U(y) - \nabla_y \log \nu(y)$$

carries the positional gradient of the pushforward density at every MCMC step, the term avoided by diffusing under μ above; training then feeds on a replay buffer of past AIS samples reweighted against staleness. The scheme of this paper is score-free in the flow: every chain targets μ with score $-\nabla U$, the flow enters only through evaluations of the log-ratio z , and each gradient step runs Algorithm 1 afresh on a new source batch, so no buffer is kept; the one persistent set, the quench and temper pool of Section 3.3, is built once and covered by sample robustness. FAB explores through its AIS chain, which must still cross the barriers of μ ; here mode discovery is delegated to the quench, which uses no such chain. The trade-offs are real: the order-2 Rényi divergence behind μ^2/ν carries a stronger tail-coverage incentive than the forward KL, and with sufficient AIS budget FAB needs no wide-coverage construction.

3.3 Constructing $\hat{\mu}$ by quench and temper

The mixture term needs samples from the wide-coverage measure $\hat{\mu}$. These are prepared once, before training, as a fixed pool of P samples $x_1, \dots, x_P$ distributed as $\hat{\mu}$; at each training step a mini-batch is drawn from

this set and refreshed by a short Langevin chain before entering the X_ω computation of Section 3.1 with μ replaced by $\hat{\mu}$. The set is produced by quench and temper (QT), a map

$$\hat{\mathcal{Y}} = \text{QT}(U)$$

from the target potential U to a wide-coverage set $\hat{\mathcal{Y}} = \{x_1, \dots, x_P\}$ distributed as $\hat{\mu}$, proceeding in three stages: diffusion (melt), optimization (quench), and rejuvenation (temper).

Algorithm 2: Quench and temper (QT): proactive mode discovery

Input: target potential U (with fixed pool size P and diffusion scale σ_{QT})
Output: wide-coverage set $\hat{\mathcal{Y}} = \{x_1, \dots, x_P\}$ forming $\hat{\mu}$, written $\hat{\mathcal{Y}} = \text{QT}(U)$

- 1 Draw an initial pool $x_1, \dots, x_P \sim \mu_0$;
// diffusion (melt)
- 2 Scatter the samples to a wider region of $\mathbb{R}^d$, $x_i \leftarrow x_i + \sigma_{\text{QT}} \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, I_d)$ i.i.d. and large $\sigma_{\text{QT}} > 0$;
// optimization (quench)
- 3 Drive each sample toward a mode center of μ by L-BFGS on U , updating $x_1, \dots, x_P$;
// rejuvenation (temper)
- 4 Spread the samples around each mode, $x_{1:P} \leftarrow \text{langevin}(x_{1:P}, U)$;

The output samples concentrate near the mode centers of μ , and the initial Brownian diffusion ensures that even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ on which the current pushforward ν places too little mass, driving the flow to discover modes that AIS alone could never reach.

A large diffusion σ_{QT} biases the QT output toward modes that lie far from the bulk of μ_0 . This is desirable rather than problematic: the modes the current pushforward fails to cover are those far from μ_0 , and the bias is also harmless because the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$.

Under typical conditions the wide-coverage set $\hat{\mathcal{Y}}$ lies within the support of μ , and the temper step samples it directly. On extreme targets — heavy tails or stiff potentials — the L-BFGS quench can instead drive particles outside the support, into high-energy regions whose weights then destabilize training (the Langevin temper itself, like the AIS rejuvenation, stays well behaved). A simple remedy, a regularization we call δ -reweighted QT, tempers the rejuvenation: run the temper step on the flattened potential $(1 - \delta)U$,

$$x_{1:P} \leftarrow \text{langevin}(x_{1:P}, (1 - \delta)U), \quad \delta \in (0, 1),$$

whose stationary law $\propto e^{-(1-\delta)U}$ is broader and integrates stably, and attach to each sample the self-normalized importance weight $w_i = e^{-\delta U(x_i)} / \sum_j e^{-\delta U(x_j)}$, which keeps $\hat{\mu}$ a probability distribution. The weighted samples then target $e^{-(1-\delta)U} \cdot e^{-\delta U} = e^{-U}$, so the broadened set keeps its wide coverage while the weights downweight the leaked, high-energy samples in the $X_{\hat{\mu}}$ average — a *statistical* removal of the bad samples rather than a hard cutoff. By sample robustness the minimizer is unchanged; the tempering only improves the conditioning of the wide-coverage construction. We write this δ -reweighted construction as

$$\hat{\mathcal{Y}} = \text{QT}_\delta(U),$$

recovering plain $\text{QT}(U) = \text{QT}_0(U)$ at $\delta = 0$.

3.4 Importance sampling at evaluation: ESS and coverage

At evaluation time the trained flow G supplies the model distribution $\nu = (G^{-1})_\# \mu_0$ as an approximation to μ , together with the exact tractable density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$ that any normalizing flow makes available. Because the density is exact, the importance weight $w(y) = \mu(y)/\nu(y) = \exp(z(y))$ of (11) can be

evaluated for free on any pushforward sample and used at test time to correct the residual bias of ν as an approximation to μ . Drawing $\mathbf{y} = (y_1, \dots, y_N)$ with $y_i = G^{-1}(x_i)$ and $x_i \sim \mu_0$ i.i.d., the weights $w(y_i)$ are put to two uses. First, they diagnose how well ν approximates μ , through the effective sample size defined below. Second, they turn the pushforward into approximate samples of μ : one resamples $\mathbf{y}$ by the weights and refreshes with a short Langevin chain targeting μ , the surrogate that drove the AIS step in Section 3.2 but now applied with the trained flow held fixed. The training bias the X functionals were designed to reduce is, at this stage, traded for sampling cost: the better the flow, the more uniform the weights and the less reweighting the resampling has to do.

We assess the quality of this pipeline through two complementary metrics: the effective sample size, which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ has been reached at all.

The first metric is the effective sample size, normalized to $[0, 1]$ by dividing the classical ESS by the sample count N . Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from ν , with importance weights $w(y_i)$ given by (11). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{\left(\sum_{i=1}^N w(y_i)\right)^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (13)$$

where the upper bound 1 follows from the Cauchy-Schwarz inequality and is attained when all weights are equal. The factor N in the denominator distinguishes our normalized ESS from the classical effective sample size $(\sum_i w(y_i))^2 / \sum_i w(y_i)^2 \in [1, N]$ standard in the literature (Elvira et al., 2022): that quantity counts an effective number of samples and scales with the batch, whereas dividing by N leaves the per-sample efficiency, a fraction in $[0, 1]$ comparable across batch sizes. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support* of ν , but modes of μ that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS, which gives rise to the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of Naeem et al. (2020). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}$ produced by the quench and temper algorithm (Algorithm 2). For each $1 \leq i \leq P$, the κ -nearest-neighbor radius of x_i (with $\kappa = 5$) is

$$\text{NND}_\kappa(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq \kappa \right\}, \quad (14)$$

i.e., the distance from x_i to its κ -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_\kappa(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1}_{\left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_\kappa(x_i; \mathbf{x}) \right\}}, \quad (15)$$

i.e., the fraction of points in $\mathbf{x}$ whose κ -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}$ has been reached by the flow, while a coverage substantially below 1 exposes the missing modes that the ESS alone cannot see. Because $\hat{\mu}$ is the same wide-coverage measure that $X_{\hat{\mu}}$ trains toward, the coverage metric certifies reach of the QT-discovered modes only, and inherits QT’s blind spot for thin heavy tails, mitigated by the tempered rejuvenation of Section 3.3.

4 Adaptive-temperature Boltzmann generator

Section 3 introduced the X-regularized forward KL (10), which improves the sample quality of the bare forward KL. In high dimensions — molecular potentials being the prime example — training a single flow map directly from the source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$ is infeasible, however good the loss. The standard remedy in Boltzmann generators is to decompose this one hard problem into a sequence of easier ones along a temperature ladder, so that each flow need only learn a single stage — a small change in the potential. This section shows how to build that ladder adaptively from the ESS metric.

4.1 Construction of temperature ladder

To bridge the simple source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$, we seek an increasing sequence

$$0 = t_0 < t_1 < \dots < t_K = 1$$

defining a sequence of intermediate distributions

$$\mu_k \propto \exp(-U_{t_k}), \quad U_{t_k}(x) = (1 - t_k) U_0(x) + t_k U_1(x), \quad k = 0, 1, \dots, K,$$

with $U_1 := U$ an alias for the target potential, so that μ_0 is the source and μ_K the target, together with K flow maps $\{G_k\}_{k=1}^K$ trained so that $(G_k^{-1})_{\#} \mu_{k-1} \approx \mu_k$. The number of stages K is left flexible; the crucial design choice is each increment $\Delta t_k = t_k - t_{k-1}$. Too large an increment degrades the training quality of stage k , while too small an increment wastes computation, so we want the largest step that each flow can still learn well.

Training the flow G_k requires samples from its target μ_k , which we obtain by SMC (Algorithm 1 with G the identity). The smallest per-step ESS of that SMC run is then a natural criterion for the step Δt_k : it measures how well the run bridges μ_{k-1} to μ_k , so we enlarge the step until this ESS drops to a fixed floor.

Algorithm 3: Adaptive selection of the temperature step Δt_k

Input: potentials U_0, U , previous level t_{k-1} , selection pool $\mathcal{P}$ of P samples from μ_{k-1} , floor $\tau = 0.7$, shrink factor $\gamma = 0.7$, initial guess t_{init}

Output: next level t_k , giving the step $\Delta t_k = t_k - t_{k-1}$

```

1  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U_1$ ;
   // start from the supplied initial guess
2  $t_k \leftarrow t_{\text{init}}$ ;
3 while true do
   // run classical SMC across the candidate step
4   Run  $M$ -step SMC on the pool  $\mathcal{P}$  (Algorithm 1 with  $G$  the identity,  $M$  as in Remark 2) from the
     distributions  $\mu_{k-1} \propto e^{-U_{t_{k-1}}}$  to  $\mu_k \propto e^{-U_{t_k}}$ , where the potential  $U_{t_k} = (1 - t_k)U_0 + t_k U_1$ ;
   // score the step by its worst per-step ESS
5   Record  $\text{ESS}_{\min}$ , the smallest per-step ESS over the  $M$  steps;
6   if  $\text{ESS}_{\min} \geq \tau$  then
7     | return  $t_k$ ;
8   else
9     | // shrink the step toward  $t_{k-1}$ 
9     |  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ ;
10  end
11 end
```

These SMC runs are classical, involving only the potentials U_0 and U and never the flow map, and are therefore cheap. They also use no QT samples, so the per-step ESS they report may be a fake ESS that overlooks modes the SMC chain never reaches. This is harmless here, because Algorithm 3 selects only the increment Δt_k and never produces training samples, so its ESS is used solely to gauge the shape deformation $\mu_{k-1} \rightarrow \mu_k$ and needs no faithful target. The floor $\tau = 0.7$ and shrink factor $\gamma = 0.7$ are empirical; the initial guess t_{init} is supplied by the outer loop (Algorithm 4). The procedure returns the largest admissible step below the initial guess: the coarsest level t_k , and hence the most distant intermediate U_{t_k} , on which the flow G_k can still be trained reliably from $U_{t_{k-1}}$. It always terminates, since as $t_k \rightarrow t_{k-1}$ the bridge $\mu_{k-1} \rightarrow \mu_k$ collapses and $\text{ESS}_{\min} \rightarrow 1 \geq \tau$, so the shrink loop is exited after finitely many steps.

4.2 The full training procedure

The two ingredients now assemble into a single loop (Algorithm 4) that grows the ladder one stage at a time, from U_0 to U , carrying a validation set $\mathcal{Y}_k \approx \mu_k$ across stages. Stage k runs five steps, with steps (iii)–(v) wrapped in an acceptance loop that shrinks the increment and retrains whenever the trained flow’s

validation ESS falls below a floor τ_v ; the outer loop stops once t_k reaches 1, so the number of stages K is selected automatically.

- (i) *Source samples.* Stage k consumes source samples in three ways, governed by two size parameters: the *pool size* P and the *batch size* B . First, a selection pool of P samples, drawn from the validation set $\mathcal{Y}_{k-1} \approx \mu_{k-1}$ carried over from the previous stage and refreshed by a short Langevin rejuvenation on $U_{t_{k-1}}$, supplies the SMC particles for the adaptive step selection of step (ii). Second, the quench and temper of step (iii) melts an initial pool of P samples from μ_0 (Algorithm 2). Third, the training batches of step (iv) are drawn at run time, B samples per gradient step, from the same $\mathcal{Y}_{k-1}$; no separate training set is constructed.
- (ii) *Adaptive step.* The next level t_k is fixed by the adaptive selection scheme (Algorithm 3): starting from an initial guess t_{init} and shrinking, it takes the largest step below the guess whose classical SMC ESS from μ_{k-1} to μ_k , run on the selection pool of step (i), stays above the floor τ . The first stage uses the safe start $t_{\text{init}} = t_{\text{safe}}$: the leading increment faces the largest deformation, and an over-aggressive start is exposed only after a full training, so the first stage begins from a level it is sure to handle. Later stages extrapolate the accepted increments, $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ with enlarge factor Γ : the step sizes are allowed to grow, since only the first few stages are difficult. This sets the stage target $\mu_k \propto \exp(-U_{t_k})$ with $U_{t_k} = (1 - t_k)U_0 + t_k U_1$.
- (iii) *Wide-coverage set.* A wide-coverage set $\hat{\mathcal{Y}}_k$ of P samples, distributed as $\hat{\mu}_k$, is drawn by quench and temper (Algorithm 2) on the stage target U_{t_k} , spreading samples across its modes. It supplies the wide-coverage half of the mixture term in the training loss.
- (iv) *Training loss.* The flow G_k is trained by the X-regularized forward KL of Section 3.1, specialized to the stage $\mu_{k-1} \rightarrow \mu_k$ at the balanced hyperparameters $\lambda = 1$, $\alpha = \beta = \frac{1}{2}$. With model $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ and its stop-gradient copy $\bar{\nu}_k$, the stage log-ratio is

$$z_k(y) = U_{t_{k-1}}(G_k(y)) - U_{t_k}(y) - \log |\det J_{G_k}(y)|,$$

and the loss reads

$$\mathcal{L}[G_k] = \underbrace{\mathbb{E}_{y \sim \mu_k} [z_k(y)]}_{\text{KL}(\mu_k \| \nu_k)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu_k} [|z_k(y) - z_k(y')|]}_{\text{X}_{\mu_k}(\mu_k \| \nu_k)} + \underbrace{\mathbb{E}_{y, y' \sim (\hat{\mu}_k + \bar{\nu}_k)/2} [|z_k(y) - z_k(y')|]}_{\text{X}_{(\hat{\mu}_k + \bar{\nu}_k)/2}(\mu_k \| \nu_k)}, \quad (16)$$

its μ_k -expectations estimated by AIS: the level- $(k-1)$ batch $\mathcal{X}$ is pushed to $\bar{\mathcal{X}}_\nu = G_k^{-1}(\mathcal{X})$, distributed as ν_k , and annealed to μ_k by Algorithm 1 with $G = G_k$, sharing the ladder length M and rejuvenation parameters with the SMC of step (ii); the wide-coverage measure $\hat{\mu}_k$ is realized by the set $\hat{\mathcal{Y}}_k$ of step (iii).

- (v) *Validation set update.* Once G_k is trained, the validation set is advanced from $\mathcal{Y}_{k-1}$ to $\mathcal{Y}_k \approx \mu_k$ by importance sampling. Each $y \in \mathcal{Y}_{k-1}$ is pushed through the trained inverse to $\tilde{y} = G_k^{-1}(y)$, a sample of $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$, and carries the importance weight

$$w_k(\tilde{y}) = \frac{\mu_k(\tilde{y})}{\nu_k(\tilde{y})} = \exp(z_k(\tilde{y})).$$

Its effective sample size $\text{ESS}(w_k)$ is the final, flow-based check on the increment: the adaptive selection of step (ii) sizes t_k from a classical SMC ESS that uses no QT samples and may therefore report a fake ESS, whereas $\text{ESS}(w_k)$ measures the trained flow directly. If $\text{ESS}(w_k) < \tau_v$ (we use $\tau_v = 0.3$), the increment was too aggressive: we discard G_k , shrink it to $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$, and repeat steps (iii)–(v) at the reduced step. Otherwise the validation set is advanced to $\mathcal{Y}_k = \text{langevin}(\text{resample}(\hat{\mathcal{Y}}, w_k), U_{t_k}) \approx \mu_k$: resample $\hat{\mathcal{Y}}$ by its weights w_k , then rejuvenate with a short Langevin chain on U_{t_k} .

The trained ladder is a Boltzmann generator for μ : starting from $x \sim \mu_0$, each stage applies its map G_k^{-1} and then reweights and resamples by the importance weights of Section 3.4, exactly as in step (v), carrying the samples from the source to the target one bridge at a time.

Algorithm 4: Adaptive-temperature Boltzmann generator training

Input: potentials U_0, U , pool size P , batch size B , validation ESS floor $\tau_v = 0.3$, shrink factor $\gamma = 0.7$,
enlarge factor $\Gamma = 2$, safe start $t_{\text{safe}} = 0.2$

Output: ladder $0 = t_0 < \dots < t_K = 1$ and flow maps $\{G_k\}_{k=1}^K$ with $(G_k^{-1})_{\#} \mu_{k-1} \approx \mu_k$

// validation set, $\mathcal{Y}_k \approx \mu_k$

```
1  $\mathcal{Y}_0 \leftarrow$  samples from  $\mu_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Form the stage source potential,  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U_1$ ;  
   // (i) source samples (selection pool)  
5   Draw a pool  $\mathcal{P}$  of  $P$  samples from  $\mathcal{Y}_{k-1}$  and rejuvenate,  $\mathcal{P} \leftarrow \text{langevin}(\mathcal{P}, U_{t_{k-1}})$ ;  
   // (ii) adaptive temperature selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by adaptive selection (Algorithm 3) on the pool  $\mathcal{P}$  with initial guess  $t_{\text{init}}$ ;  
12  Form the target potential  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU_1$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) wide-coverage set  
15    Draw the wide-coverage set of  $P$  samples by quench and temper,  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(U_{t_k})$  (Algorithm 2);  
    // (iv) X-regularized training loss  
16    for each gradient step do  
17      Draw a training batch of  $B$  samples  $\mathcal{X}$  from the validation set  $\mathcal{Y}_{k-1}$ ;  
18      Form the detached model batch,  $\tilde{\mathcal{X}}_\nu \leftarrow G_k^{-1}(\mathcal{X})$ , distributed as  $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ ;  
19      Anneal from  $\tilde{\mathcal{X}}_\nu$  to  $\mathcal{X}_\mu$  via AIS (Algorithm 1, from  $\nu_k$  to  $\mu_k$ );  
20      Draw a wide-coverage batch  $\hat{\mathcal{X}}_\mu \subset \hat{\mathcal{Y}}_k$  and rejuvenate,  $\hat{\mathcal{X}}_\mu \leftarrow \text{langevin}(\hat{\mathcal{X}}_\mu, U_{t_k})$ ;  
21      Update  $G_k$  by a gradient step on the X-regularized forward KL (16), from  $\mathcal{X}_\mu$  and the balanced  
      mixture of  $\hat{\mathcal{X}}_\mu$  and  $\tilde{\mathcal{X}}_\nu$ ;  
22    end  
    // (v) validation set update  
23    Push the validation set  $\tilde{\mathcal{Y}} \leftarrow G_k^{-1}(\mathcal{Y}_{k-1})$ ;  
24    Compute the weights  $w_k \leftarrow \exp(U_{t_{k-1}}(\mathcal{Y}_{k-1}) - U_{t_k}(\tilde{\mathcal{Y}}) - \log |\det J_{G_k}(\tilde{\mathcal{Y}})|)$ ;  
25    if  $\text{ESS}(w_k) < \tau_v$  then  
26      // abort the step and shrink toward  $t_{k-1}$   
       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then re-form  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU_1$ ;  
27    else  
28      Resample  $\mathcal{Y}_k \leftarrow \text{resample}(\tilde{\mathcal{Y}}, w_k)$ ;  
29      Employ Langevin rejuvenation to obtain  $\mathcal{Y}_k \leftarrow \text{langevin}(\mathcal{Y}_k, U_{t_k})$ ;  
30    end  
31  until  $\text{ESS}(w_k) \geq \tau_v$ ;  
32 end  
33  $K \leftarrow k$ ;  
34 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

In particular, setting every stage flow G_k to the identity reduces the adaptive-temperature generator to annealed SMC, a purely Monte Carlo sampler that requires no training and serves as the benchmark against the flow-based methods. With no flow to train, steps (i), (iii), and (iv) (the source pool, the wide-coverage set, and the X-regularized loss) fall away, leaving only step (ii), the adaptive temperature selection, and step (v), the reweight, resample, and rejuvenation. The pushforward is then $\hat{\mathcal{Y}} = \mathcal{Y}_{k-1}$ and the log-Jacobian vanishes, so the stage weight reduces to the classical tempering weight $w_k = \exp(U_{t_{k-1}}(\mathcal{Y}_{k-1}) - U_{t_k}(\mathcal{Y}_{k-1}))$, the outer-ladder counterpart of the identity-flow reduction of Algorithm 1. Run on the same adaptive ladder and acceptance standard, it isolates the value the trained flow adds over annealing alone.

5 Numerical experiments on model problems

We validate the X-regularized losses on a suite of model targets; the application to a real molecular system is taken up in Section 6. The suite spans two families: multimodal probability distributions, the four 2D benchmarks (Section 5.1) and the high-dimensional product multi-well (Section 5.2); and lattice field theories, the tilted ϕ^4 theory (Section 5.3) and the p -state clock model (Section 5.4). Throughout this section, vectors are set in bold ($\mathbf{x}$, $\mathbf{y}$, $\boldsymbol{\theta}$, $\boldsymbol{\phi}$) and their scalar components and scalar quantities upright.

5.1 2D benchmark distributions

We test the training scheme on four 2D targets, each chosen to isolate a different failure mode of the forward KL rather than to match the Gaussian mixture and double-well community suite (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024): Threewell, Himmelblau, Annulus, and Sparse. The potentials and mode structures are listed in Table 2; each target’s construction is described alongside its results below.

name	potential $U(x_1, x_2)$	minima
Threewell	$6[(x_1^2 - 1)^2 + (x_2^2 - 1)^2 + \sin(x_1 + 2x_2)]$	three asymmetric wells
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{\frac{1}{3}}, r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$
Sparse	Sparse Gaussian mixture	2 dense + 2 isolated centers

Table 2: Four 2D benchmark targets.

For all four targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$ with σ between 0.25 and 2.0; on Threewell it is deliberately narrow and centered at the origin, far from all three wells, so the initial pushforward covers none of them. The flow is a neural spline flow (NSF) (Durkan et al., 2019) with 6 coupling transforms, 32 bins, and hidden widths (128, 128), on a square box matched to each target. Training uses Adam at learning rates 10^{-3} to 5×10^{-3} for 1000 steps, with batch size 500, enlarged on the multimodal Threewell and Sparse so a single batch can cover all modes. At each step the AIS surrogate is one importance-weighted resampling pass against the current pushforward followed by 50 Langevin steps. The loss coefficients are the balanced hyperparameters of Section 3.1, $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$, and the wide-coverage measure $\hat{\mu}$ is built once before training by QT (Algorithm 2) and recycled with one Langevin rejuvenation per step.

For each target we compare four losses obtained by progressively adding one X functional at a time: the bare forward KL (2); $\text{KL} + \text{X}_{\mu}$ with $\lambda = 1$; $\text{KL} + \text{X}_{\mu} + \text{X}_{\hat{\mu}}$ with $\alpha = 1, \beta = 0$; and the balanced mixture $\text{KL} + \text{X}_{\mu} + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ with $\alpha = \beta = \frac{1}{2}$.

Threewell. The quartic-plus-sinusoid potential has three asymmetric wells. Forward KL and $\text{KL} + \text{X}_{\mu}$ lock onto only two of them and never discover the third, yet their ESS stays close to one: a fake ESS instance, the flow omitting a mode while reporting near-uniform weights on the support it reaches. Adding $\text{X}_{\hat{\mu}}$ places mass on the QT-discovered third well, and the mixture attains the same full coverage at the highest full-coverage ESS (Figure 1a).

Himmelblau. Forward KL and KL+ X_μ both cover only two of the four modes, yet the ESS gives no warning: KL+ X_μ posts the highest nominal ESS of the four while missing half the modes, a fake ESS instance. Adding $X_{\hat{\mu}}$ or the mixture recovers full coverage and ties at the best full-coverage ESS (Figure 1b).

Annulus. All four methods reach full coverage, so the discriminator is the flow shape and ESS. Forward KL alone leaves visible axis-aligned filaments across the central barrier; adding X_μ tightens the pushforward to a clean ring-plus-point geometry and sharply lifts the ESS, and the mixture ties KL+ X_μ at the best ESS of the four (Figure 2a). This isolates the role of X_μ as a shape regularizer, penalizing pointwise log-ratio variation that the global forward KL leaves untouched.

Sparse. A four-mode Gaussian mixture with a dense central pair and two isolated corner modes the narrow source cannot reach. Forward KL and KL+ X_μ lock onto the central pair and miss the corners at high, stable ESS, another fake ESS instance. Adding $X_{\hat{\mu}}$ restores full coverage but, lacking the $\bar{\nu}$ half, overfits $\hat{\mu}$ and leaves artificial bridges of mass between the modes; the mixture also reaches full coverage while removing these bridges, at the best full-coverage ESS (Figure 3).

target	KL	KL+ X_μ	KL+ X_μ + $X_{\hat{\mu}}$	KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$
Threewell	0.99 (0.75)	1.00 (0.75)	0.98 (1.00)	0.99 (1.00)
Himmelblau	0.92 (0.53)	0.97 (0.48)	0.95 (1.00)	0.95 (1.00)
Annulus	0.66 (1.00)	0.97 (1.00)	0.94 (1.00)	0.97 (1.00)
Sparse	0.88 (0.50)	0.94 (0.50)	0.87 (1.00)	0.93 (1.00)

Table 3: Final ESS, with $\kappa = 5$ coverage against $\hat{\mu}$ in parentheses, for each loss on each 2D target; the best full-coverage ESS per row is in bold.

Across the four targets two regimes emerge (Table 3). On the mode discovery targets Threewell, Himmelblau, and Sparse, the ESS stays high for every loss and is therefore useless on its own: forward KL and KL+ X_μ post a high ESS while silently missing modes, and only the $X_{\hat{\mu}}$ and mixture variants restore full coverage. On the shape and manifold target Annulus, coverage is already complete and ESS becomes the discriminator, where X_μ helps and the mixture attains the best ESS. Overall the mixture is the only loss of the four that reaches full mode coverage and the best or tied-best full-coverage ESS on every target, and its ESS curves are smoother than bare forward KL.

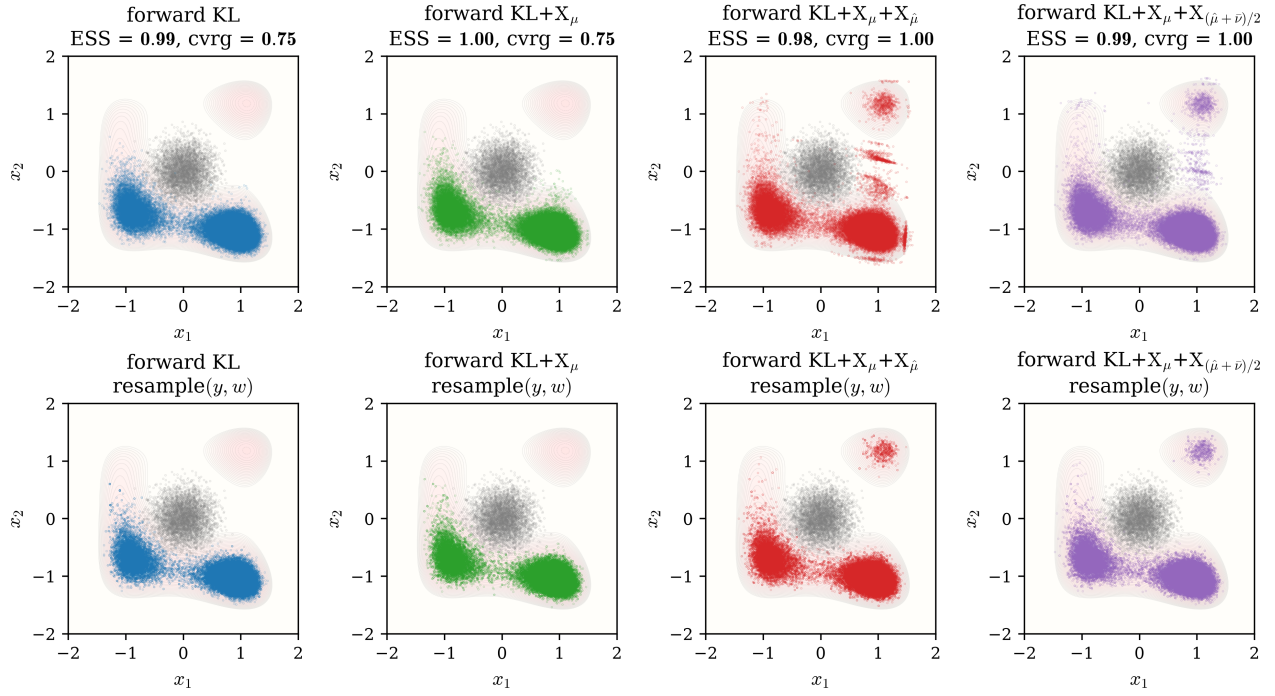
The mixture loss $\mathcal{L} = \text{KL}(\mu\|\nu) + X_\mu(\mu\|\nu) + X_{(\hat{\mu}+\bar{\nu})/2}(\mu\|\nu)$ with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$ is the recommended default. It absorbs the mode discovery role of $X_{\hat{\mu}}$ while suppressing the leakage that $X_{\hat{\mu}}$ alone can introduce when the target has holes interleaved with its modes.

5.2 High-dimensional product multi-well

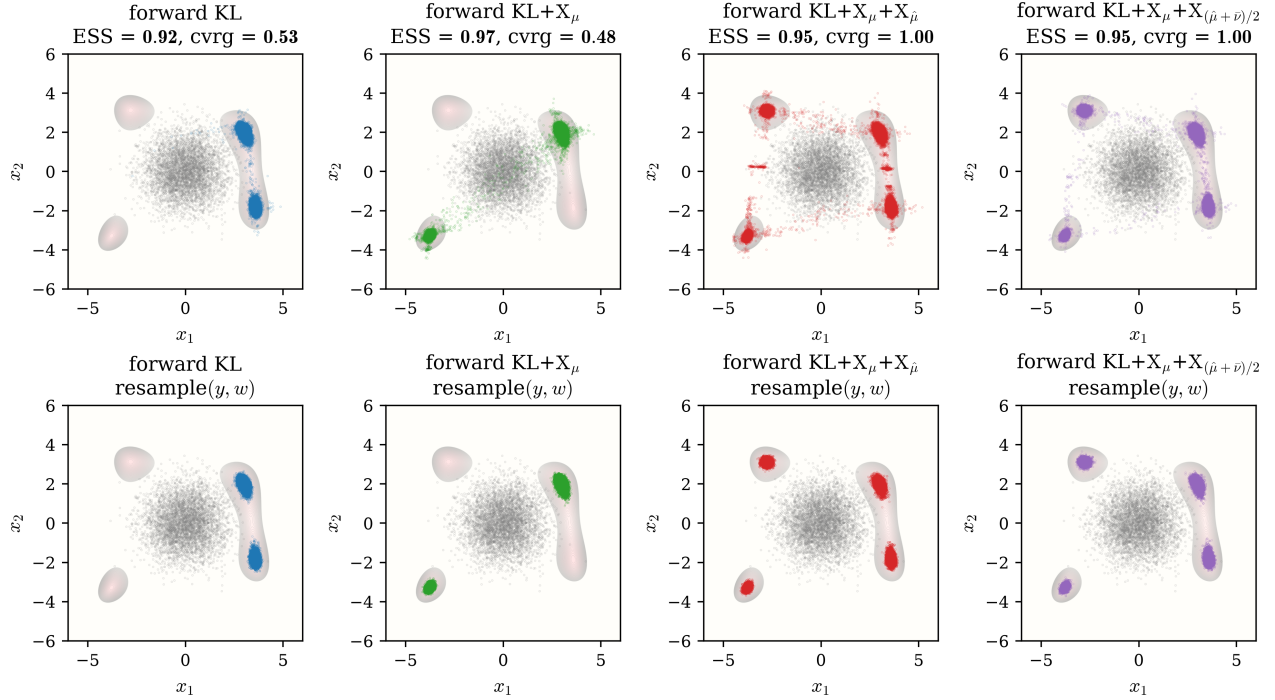
To test the framework beyond two dimensions we construct a family of product multi-well targets in dimension $d = 2^k$, playing here the role that double-well systems play in the diffusion-sampler literature (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024), but with an analytically known mode structure and a diagonal optimal transport map so that training failures are unambiguously training failures and not capacity failures of the flow,

$$U(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^{2^k} x_i^2 + 12 \sum_{i=1}^k \exp(-x_i^2), \quad (17)$$

a double well in each of the first k coordinates, with minima at $\pm\sqrt{\ln 24} \approx \pm 1.78$ separated by a barrier of height ≈ 9.9 , and a standard Gaussian in the remaining $2^k - k$. The source is the standard Gaussian $\mu_0 = \mathcal{N}(0, I_d)$, whose potential is the quadratic part of U . Because U is separable, the target factorizes into k identical double wells and $2^k - k$ Gaussians, so it has exactly $2^k = d$ modes, and the optimal transport map is diagonal. The source sits exactly on the barrier of every double well, so the untrained pushforward covers none of the 2^k mode combinations. We sweep $k = 1, \dots, 8$, i.e. $d = 2, 4, 8, 16, 32, 64, 128, 256$.



(a) Threewell target.



(b) Himmelblau target.

Figure 1: Mode discovery targets. In each panel, Row 1 is the pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$ and Row 2 is $\text{resample}(\mathbf{y}, w)$ with Langevin rejuvenation. Columns left to right: forward KL, KL+ X_{μ} , KL+ $X_{\mu}+X_{\hat{\mu}}$, KL+ $X_{\mu}+X_{(\hat{\mu}+\hat{\nu})/2}$. Each pushforward title reports the final ESS and the $\kappa = 5$ coverage against $\hat{\mu}$.

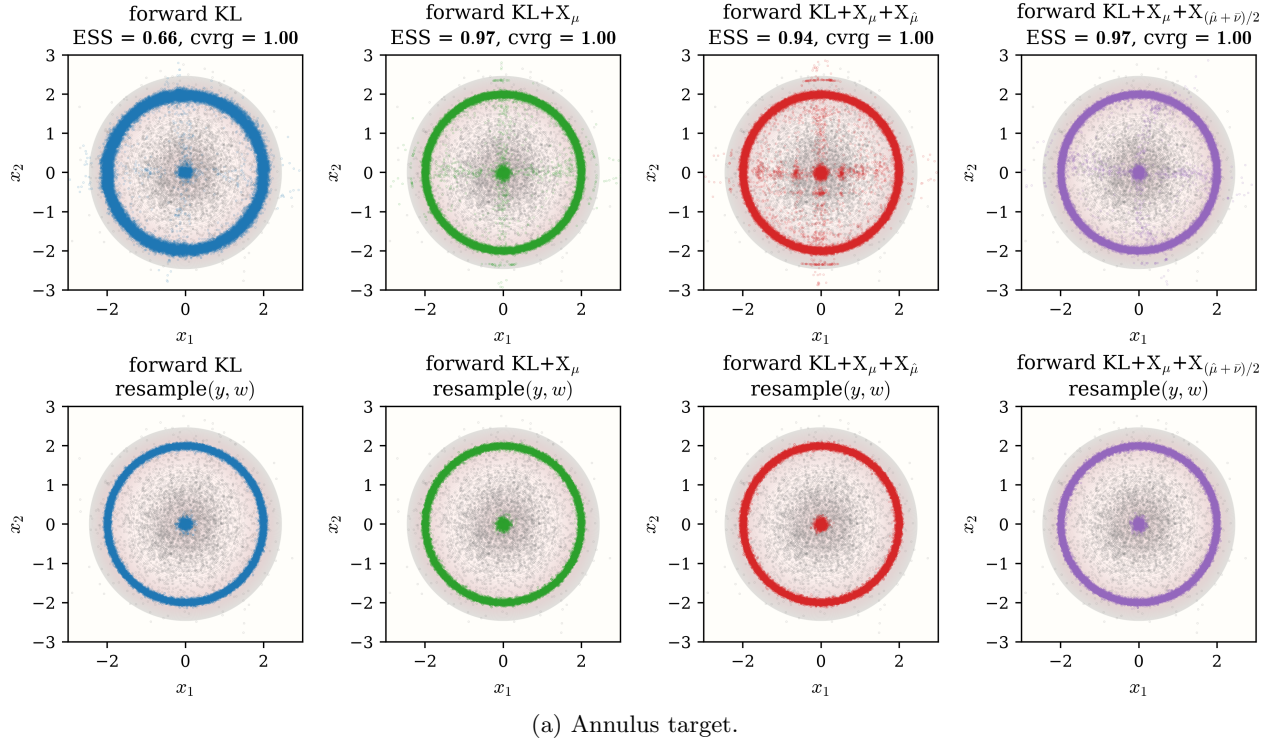


Figure 2: Annulus target. Rows and columns as in Figure 1.

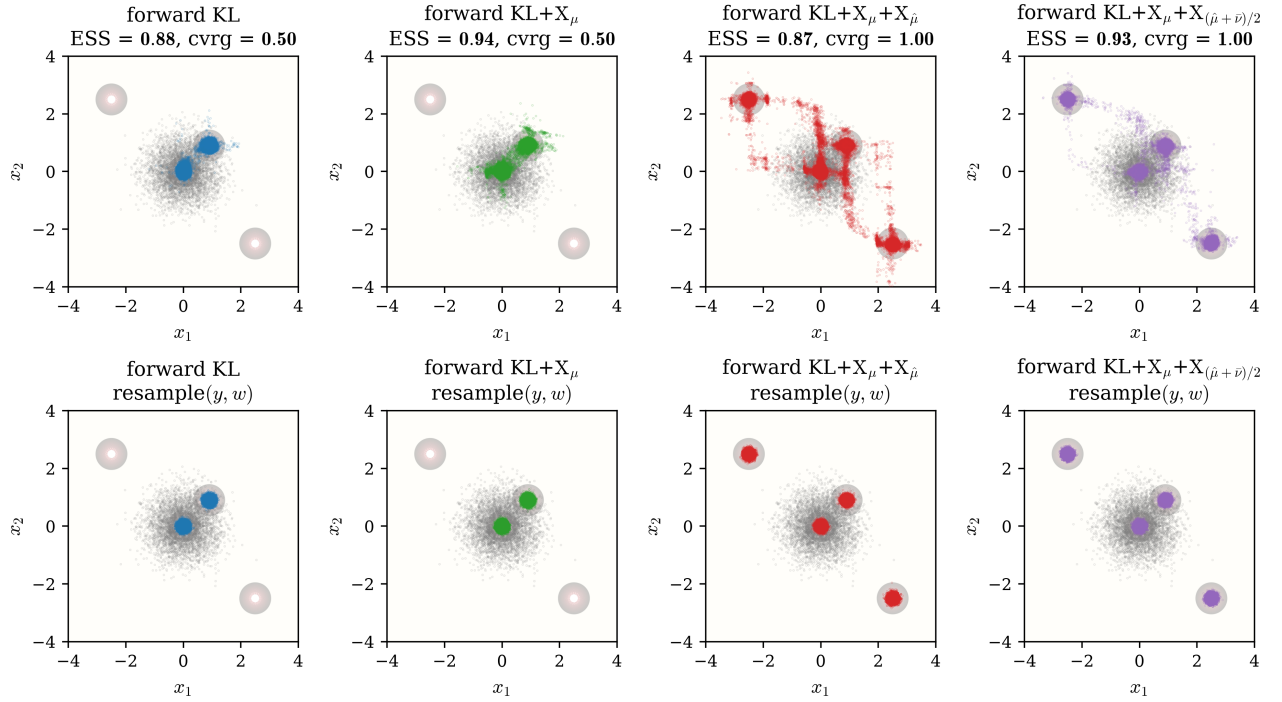


Figure 3: Sparse target. Row 1: pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$. Row 2: resample($\mathbf{y}, w$) with Langevin rejuvenation. Columns as in Figure 1.

The flow is a neural spline flow on $[-4, 4]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). Training uses Adam at learning rate 10^{-3} with batch size 200, drawn from a source set of $10^4 \cdot 2^k$ samples; forward KL is sensitive to the batch size, so the small batch sharpens the contrast between the losses. The number of steps decreases with d from 3000 at $d \leq 4$ to 1200 at $d \geq 64$. The μ -surrogate is the same one importance-weighted resampling pass followed by 50 Langevin steps used in the 2D experiments, and $\hat{\mu}$ is built once by QT from $100 \cdot 2^k$ samples. The four losses are those of Section 5.1, with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$.

Table 4 reports the final ESS of each loss across the eight dimensions. Unlike the 2D mode discovery targets, the flow reaches all 2^k modes for every loss, each mode filled to within a few percent of its uniform 2^{-k} share and the $\kappa = 5$ coverage against $\hat{\mu}$ at least 0.94 throughout, so this is the calibration regime of Annulus rather than the fake ESS regime, and ESS is the sole discriminator. Three trends emerge. First, the bare forward KL degrades monotonically with dimension, as the log-weight variance accumulates one double well at a time. Second, the X functionals counteract this and the gap widens with dimension up to $d = 128$: negligible at $d = 2$, and at $d = 256$ the wide-coverage $X_{\hat{\mu}}$ becomes the best loss. Third, the QT-driven losses, $X_{\hat{\mu}}$ and the mixture, overtake X_{μ} alone at high dimension: X_{μ} tracks the mixture through $d = 64$ but falls below the bare KL by $d = 256$ because it reweights the live one-step IS surrogate whose pushforward is poorly formed early in training, while $X_{\hat{\mu}}$ and the mixture hold their lead because the fixed QT set keeps the flow anchored on the wells even as the log-weight variance grows.

At $d = 256$, i.e. $k = 8$, although the nominal dimension is 256 the *essential* dimension of the target is only $k = 8$: the potential (17) is a double well in the first 8 coordinates and a standard Gaussian in the remaining 248, so the multimodality lives in an 8-dimensional subspace, while the log-weight variance, and hence the ESS, still loads on all 256. Here we train a single flow and sweep the AIS surrogate ladder length $M \in \{1, 2, 4, 8\}$ that builds the μ -surrogate for the mixture loss, the inner ladder of Algorithm 1, distinct from the temperature ladder of Section 4. Increasing M accelerates convergence in the first training stage, leaving the low ESS plateau sooner, but does not affect the long-time limit: the ESS curves differ only in the first stage and settle at a comparable value (Figure 4). The ladder length is therefore a first-stage accelerator rather than a determinant of the final flow quality.

loss \ d	2	4	8	16	32	64	128	256
forward KL	0.958	0.937	0.935	0.871	0.790	0.669	0.547	0.384
KL+ X_{μ}	0.998	0.983	0.972	0.961	0.893	0.845	0.657	0.333
KL+ X_{μ} + $X_{\hat{\mu}}$	0.987	0.990	0.973	0.955	0.948	0.835	0.750	0.565
KL+ X_{μ} + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.986	0.992	0.975	0.969	0.953	0.878	0.767	0.543

Table 4: Final ESS of each loss on the product multi-well target (17) across dimension $d = 2^k$, $k = 1, \dots, 8$; the best per column is in bold.

5.3 Lattice ϕ^4 field theory: fake ESS at a collective barrier

The scalar ϕ^4 lattice theory is a standard testbed on which flow-based samplers are known to collapse. We use its *tilted* broken phase as a sharp fake ESS instance. The target is the scalar ϕ^4 field on a periodic $L \times L$ lattice, one real value ϕ_j per site, $\phi \in \mathbb{R}^d$ with $d = L^2$, distributed as $\mu \propto e^{-U}$ with action

$$U(\phi) = -2\kappa_{\text{lat}} \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \lambda_4 (\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j,$$

with $\langle jl \rangle$ running over nearest-neighbor pairs. In the broken phase the lattice orders into two aligned states near $\pm v \mathbf{1}$, the uniform configurations of field value $\pm v$ ($\mathbf{1}$ the all-ones vector), separated by a *collective* barrier, the free energy of the domain wall any continuous path between them must build; the order parameter is the magnetization $m(\phi) = \frac{1}{d} \sum_{j=1}^d \phi_j$, whose distribution is sharply bimodal. The tilt h breaks the symmetry and carries the design: at $h = 0$ the two vacua hold weight $\frac{1}{2}$ each *by symmetry*, so a collapsed flow loses nothing that symmetrization cannot restore, and equivariant architectures bypass the failure; with

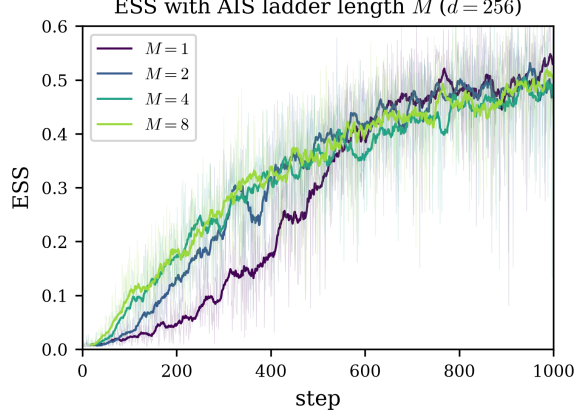


Figure 4: ESS trajectory across training steps for the mixture loss $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$ on the product multi-well at $d = 256$, as the AIS ladder length M that builds the μ -surrogate is varied over $\{1, 2, 4, 8\}$. Faint lines are the raw per-step ESS; bold lines are a moving average.

$h \neq 0$ the relative weight $p_+/p_- = e^{-\Delta F}$, with $p_+ = \mathbb{P}(m > 0)$ the weight of the minority phase, is a nontrivial number that the sampler must reproduce. We report $L = 6$ and $L = 8$. The parameters are frozen by a classical pilot scan: $\lambda_4 = \frac{1}{2}$ throughout; $\kappa_{\text{lat}} = 0.40$, $h = 0.0257$ at $L = 6$ and $\kappa_{\text{lat}} = 0.40$, $h = 0.0144$ at $L = 8$, placing the measured barrier at 8.0 and $10.2 k_B T$ and the minority-phase weight at $p_+ = 0.128$ and 0.139 . The reference is parallel tempering with the Metropolis-adjusted Langevin algorithm (MALA) over an action-scaling ladder, completing 3.7×10^3 – 4.9×10^4 ladder round trips per run.

The protocol is the single-flow pipeline of Section 5.1, unchanged: a neural spline flow on $[-3, 3]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256); source $\mathcal{N}(0, 0.25 I_d)$ centered on the barrier top; batch size 500, 2000 Adam steps at learning rate 10^{-3} ; the μ -surrogate is the one importance-weighted resampling pass followed by 50 Langevin steps; $\hat{\mu}$ is built once by QT on U , which lands in both vacua from 45–51% of melted starts. Training carries a stability guard: a non-finite loss or gradient skips the gradient step under gradient clipping. Single-flow training is deliberate: a large lattice calls for the adaptive-temperature generator of Section 4, whose hot rungs mix the vacua by construction and would dissolve the failure under study; this benchmark isolates the regime in which the ESS diagnostic itself is misleading.

	$L = 6$ ($d = 36$)		$L = 8$ ($d = 64$)	
	ESS	p_+	ESS	p_+
forward KL	0.77 ± 0.04	0, 0, 0	0.57 ± 0.09	0, 0, 1
$\text{KL}+X_\mu$	0.93 ± 0.01	0, 0, 1	0.81 ± 0.01	0, 0, 1
$\text{KL}+X_\mu+X_{\hat{\mu}}$	0.88 ± 0.01	0.126 ± 0.001	0.66 ± 0.02	0.127 ± 0.001
$\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	0.89 ± 0.01	0.124 ± 0.002	0.68 ± 0.02	0.127 ± 0.001
PT reference	—	0.128	—	0.139

Table 5: Final ESS and reweighted minority-phase weight p_+ on the tilted ϕ^4 lattice, three seeds per size; for the collapsed losses the p_+ column lists the per-seed outcomes, 1 denoting collapse onto the *minority* vacuum. Bold marks the recovered phase weights.

Table 5 and Figure 5 report the outcome. Bare forward KL and $\text{KL}+X_\mu$ collapse at every size and every seed: the AIS chain initialized at the pushforward cannot cross an 8.0 – $10.2 k_B T$ collective barrier, so the loss never sees the second vacuum, the reweighted p_+ is exactly zero or exactly one, and the estimated ΔF is infinite where the truth is finite. The collapse direction is unpredictable across seeds: at $L = 6$ one seed of $\text{KL}+X_\mu$ lands on the *minority* vacuum at a high fake ESS, and at $L = 8$ both non-QT-driven losses do so, so ΔF comes out wrong in *sign*. The fake ESS is high and reproducible for $\text{KL}+X_\mu$ at $L = 8$, and arises as follows: X_μ penalizes the spread of z over the batch, and with every batch confined to one well it sharpens

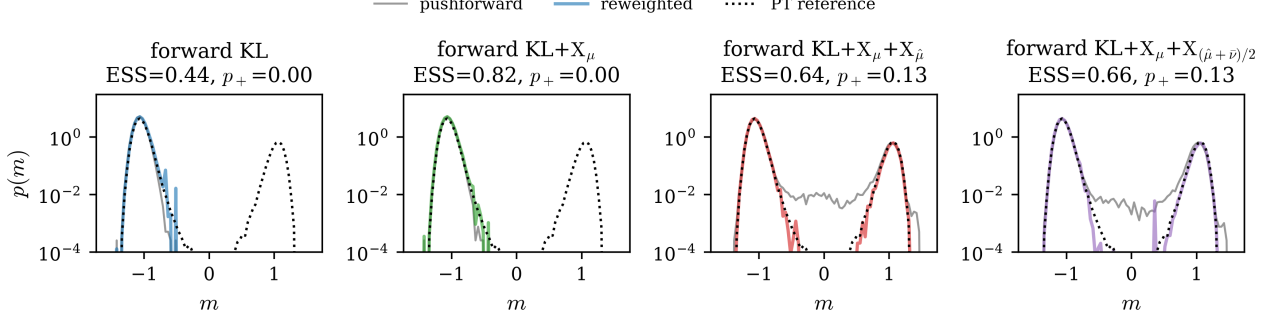


Figure 5: Magnetization density $p(m)$ on the $L = 8$ lattice, one panel per training loss. Each panel shows the pushforward density (gray), the reweighted density (color), and the PT reference (dotted); the title reports the final ESS and the reweighted p_+ .

the in-well fit, reducing the weight variance on the captured support; a variance-reducing term applied to a truncated support raises the restricted ESS while doing nothing for coverage. The ESS is thus highest where the collapse is most severe. The QT-driven losses repair both sizes at every seed: p_+ matches the reference with small seed spread, the minority well is genuinely well sampled rather than rescued by reweighting, and the cost over the collapsed runs is modest. This ESS gap has a topological reading: a diffeomorphism maps the connected Gaussian source to a connected pushforward, visible as the strictly positive gray line across the barrier in Figure 5, so any flow covering both vacua must thread mass through the inter-phase region where the target is exponentially small; that bridge carries near-zero weights, so a flow covering both vacua has a lower ESS than a collapsed one. A training signal driven by ESS alone would not favor covering both vacua; $X_{\hat{\mu}}$ forces the discovery regardless. The repaired weight tracks the action, not the QT set: retraining at $L = 6$ with QT sets deliberately skewed to 5% and 25% minority composition returns the same p_+ throughout, the sample-robustness property of X_{ω} in action. Stability-guard skips are confined to the QT-driven losses, the early-training cost of confronting cross-well log-ratio spreads.

The same pattern persists at larger lattices but weakens: at $L = 12$ the collapsed losses still discard the minority phase while their ESS falls, and the QT-driven losses still cover both phases, correct in kind but expensive and drifting in weight. The fake ESS is therefore most pronounced at moderate size, where it is high enough to mislead; beyond it, single-flow training of either loss family is no longer appropriate and the annealed generator of Section 4 takes over.

5.4 Boltzmann generator on p -state clock model

The preceding experiments train a single flow map; this experiment exercises the full adaptive-temperature procedure of Algorithm 4, contrasting the X -regularized stage loss (16) against the bare forward KL through an otherwise identical pipeline. The target is the p -state clock model on a periodic 8×8 square lattice, one angle per site,

$$U(\boldsymbol{\theta}) = -J \sum_{\langle j,l \rangle} \cos(\theta_j - \theta_l) - h \sum_{j=1}^d \cos(p\theta_j), \quad \boldsymbol{\theta} \in [-\pi, \pi]^d, \quad d = 64, \quad (18)$$

with nearest-neighbor coupling $J = 1$, $\mathbb{Z}_p$ anisotropy $h = \frac{1}{2}$, and $p = 6$ clock states; for $p \geq 5$ the model hosts two Berezinskii–Kosterlitz–Thouless (BKT) transitions enclosing a critical quasi-long-range-ordered phase (José et al., 1977). The cold measure $\propto e^{-U}$ splits into $p = 6$ equivalent sectors labeled by the phase of the global magnetization

$$m(\boldsymbol{\theta}) = \frac{1}{d} \sum_{j=1}^d e^{i\theta_j}, \quad s(\boldsymbol{\theta}) = \left\lfloor \frac{p \arg m(\boldsymbol{\theta})}{2\pi} \right\rfloor \bmod p \in \{0, \dots, 5\},$$

where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. The annealing path from the uniform measure to μ traverses the BKT window, where the correlation length grows rapidly and the six sectors deepen simultaneously. Figure 6 displays this structure on the trained validation set.

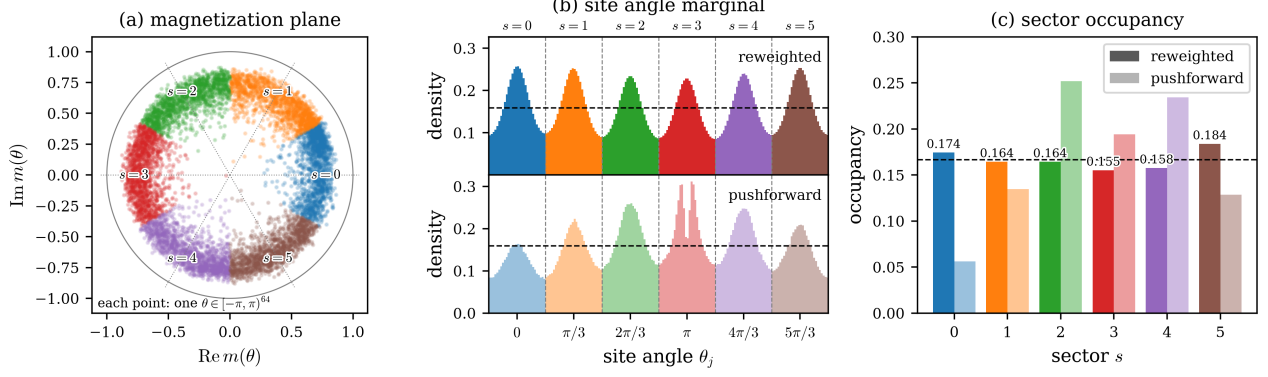


Figure 6: Structure of the cold clock measure (18) at $d = 64$, displayed on 10^6 reweighted samples: fresh source draws advanced through the trained ladder by the map–reweight–resample step (v), X-regularized loss at $B = 10^4$. (a) Global magnetization $m(\theta)$, colored by sector index $s(\theta)$, 1500 points per sector: the measure splits into $p = 6$ equivalent sectors with mean $|m| = 0.73$. (b) Site-angle marginal: the anisotropy locks each site to the six clock angles $2\pi s/6$, $s = 0, \dots, 5$; dashed lines bound the six wells, colored as in (c), and the black line marks the uniform density $\frac{1}{2\pi}$; the pushforward marginal is from the maps alone, without reweighting. (c) Sector occupancy of reweighted and pushforward samples, the black line marking the uniform occupancy $\frac{1}{6}$.

The source is the uniform measure on the torus, $U_0 = 0$, so $U_t = tU$; each stage map G_k is a circular neural spline flow on $[-\pi, \pi)^{64}$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). The protocol is deliberately large-sample: a validation set of 10^6 samples, pool size $P = 2 \times 10^5$, safe start $t_{\text{safe}} = 0.2$, shared ladder length $M = 4$, and SMC floor $\tau = 0.7$, which trims over-aggressive guesses before any training is spent. We report three batch sizes $B = 10^3, 10^4, 10^5$, with 2000, 500, 200 Adam steps per stage; the remaining constants are those of Algorithm 4. Training carries a stability guard: a non-finite loss skips the gradient step under gradient clipping and a fresh batch is drawn, with a stage abort only after repeated skips. The two losses compared are the full stage loss (16) at the balanced hyperparameters $\lambda = 1$, $\alpha = \beta = \frac{1}{2}$, and the bare forward KL, its first term alone; everything else is shared.

Table 6 reports the core metric, the validation ESS of step (v) on every accepted rung. Both losses complete the ladder at every batch size, along *identical* ladders with *identical* rejected rungs: the adaptive machinery converts loss quality into schedule, not into success or failure. Figure 7 shows the per-step training ESS on the first three stages. The discrimination lives in the per-rung quality. On accepted rungs the X-regularized loss posts a higher validation ESS on stages 1–3 at all three batch sizes; the gap closes to zero in the transition window, slightly reversing on a few late rungs. Wall time per pair is equal. A generalization check pushes 10^6 independent source samples through the trained $B = 10^4$ ladders, stage by stage with the resampling of step (v): the fresh-sample per-stage ESS tracks the accepted-rung validation ESS of Table 6 for both losses, and the final sector occupancy is uniform per sector, so the gates measure generalization rather than memorization of the carried set.

The pattern is consistent with the theory: the X terms lower the asymptotic accuracy floor of each stage (Section 2), a larger training *limit*, but promise no faster transient, and the rejected-rung values confirm that the bare forward KL trains no slower. The anatomy of the ladder shows where the gain lives. The early bridges, $t \leq 0.66$, are far but smooth: a large, structurally simple deformation with a clean AIS signal, so the training loss is the binding constraint and the X terms win by up to +0.21. In the transition window, $t \approx 0.66$ –0.95, the selector halves the accepted increment, an empirical readout of the annealing bottleneck at the BKT crossing; there the ceiling is set by the critically slow Langevin rejuvenation and the shared network capacity, constraints common to both losses, so the gap collapses to zero. The final hop to $t = 1$ is the easiest rung for both. On this target the six sectors are never isolated (Figure 6(a)), so the mode discovery channel of $X_{\hat{\mu}}$ is not load-bearing; the qualitative separations of Sections 5.1–5.2, where modes are isolated by construction, do not transfer to soft multimodality. What transfers is a net per-rung quality improvement, concentrated on the early bridges, at zero extra cost: the X-regularized loss was never worse than the bare forward KL where it mattered and never failed where the bare loss passed.

stage k	1	2	3	4	5	6	7	8
t_k ($B = 10^3$)	0.200	0.396	0.530	0.662	0.753	0.841	0.952	1.000
forward KL	0.391	0.398	0.463	0.366	0.467	0.465	0.400	0.708
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	0.602	0.524	0.559	0.401	0.500	0.453	0.369	0.778
t_k ($B = 10^4$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
forward KL	0.531	0.523	0.347	0.396	0.349	0.341	—	—
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	0.703	0.628	0.381	0.412	0.336	0.303	—	—
t_k ($B = 10^5$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
forward KL	0.611	0.602	0.416	0.461	0.400	0.387	—	—
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	0.745	0.688	0.451	0.489	0.405	0.383	—	—

Table 6: Validation ESS on every accepted rung, the acceptance gate of Algorithm 4 with floor $\tau_v = 0.3$, on the $p = 6$ clock model at $d = 64$; bold marks the better loss on each rung. Within each pair the two losses traverse the same accepted ladder and reject the same rungs, seven rejections per run at $B = 10^3$ and two at $B = 10^4$ and 10^5 .

The trained ladder is used as a staged sampler: each stage applies its map G_k^{-1} and then reweights and resamples as in step (v), and Figure 6(c) shows the resulting occupancy. The weights of step (v) are pointwise density ratios, exchangeable within a sector and hence sector-blind: they neither detect the imbalance, seeded by the slight flow bias that the i.i.d. pushforward exposes, nor correct it; the cold Langevin rejuvenation transports no mass between sectors; and the final validation set is resampled, carrying no weights at all. The occupancy is thus unbiased in expectation over runs but unrestored within any single one, and the per-stage fluctuations accumulate along the ladder. A scaling test quantifies the accumulation: rebuilding fresh sample sets of size $N = 10^4 \cdot 2^k$, $k = 0, \dots, 7$, through the trained $B = 10^4$ ladder, with 2^{7-k} independent rebuilds per size, the mean occupancy bias $\frac{1}{6} \sum_s |p_s - \frac{1}{6}|$ contracts at the fitted rate $N^{-0.34}$, shown in Figure 8, distinctly slower than the Monte Carlo rate $N^{-1/2}$ of a single multinomial draw: the staged sampler composes six map–reweight–resample steps, and each resampling correlates the surviving lineages, so the effective number of independent sector draws grows sublinearly in N . Coverage, ESS, and occupancy evenness are three distinct axes of sample quality; the failure that no post-processing could repair, a *missing* sector, does not occur for either loss.

6 Boltzmann generators on molecular potential energies

The model problems of the previous section were each chosen to isolate one failure mode of forward KL training on a smooth target: the 2D benchmarks, the high-dimensional multi-well, the lattice ϕ^4 theory, and the p -state clock model. The application that motivates a Boltzmann generator in the first place is a molecule, and a molecular target exercises the whole pipeline at once. This section applies the adaptive-temperature, X-regularized generator of the previous section to three molecules of increasing difficulty in full internal coordinates, $d = 36$ to 60, on a single consumer GPU. It is organized into five parts: the motivation and the two difficulties that make a molecular potential hard (Section 6.1); *sharpening*, the core simplification that makes the framework tractable on a singular target (Section 6.2); the molecule-specific engineering it requires (Section 6.3); the numerical results on glycerol, diethanolamine, and alanine dipeptide (Section 6.4); and the limitations of the molecular study (Section 6.5).

6.1 Motivation and background

The distribution that motivates a Boltzmann generator is a molecule: the equilibrium (Boltzmann) distribution $\mu \propto e^{-U}$ of a molecular potential, with U its dimensionless reduced form, whose samples are the conformers that set binding affinities, reaction rates, and free energies in computational chemistry and drug design (Noé et al., 2019). The targets studied so far were smooth by design; a molecular potential is not.

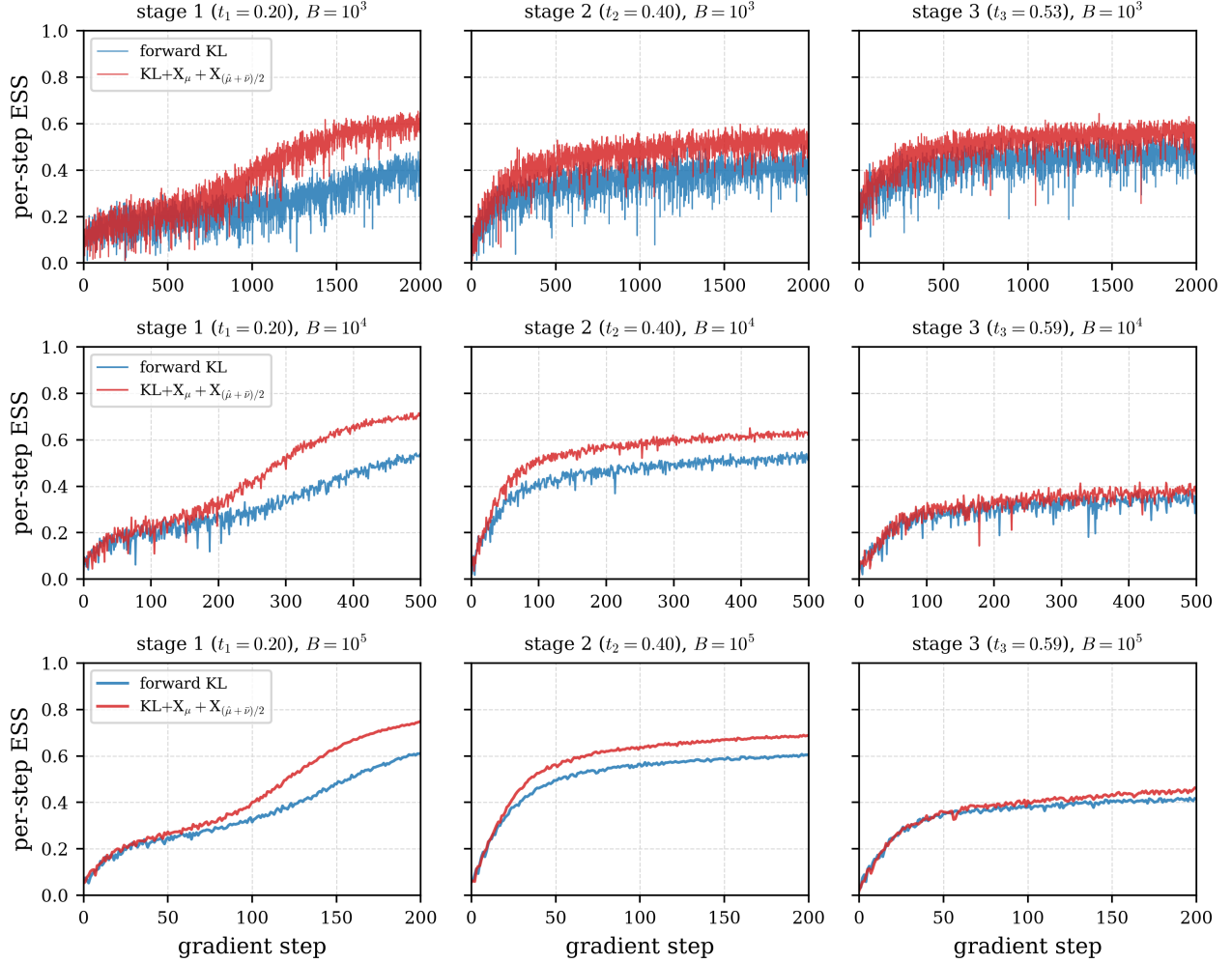


Figure 7: Per-step ESS of the training surrogate on the first three stages, bare forward KL (blue) versus the X-regularized loss (red); rows $B = 10^3, 10^4, 10^5$, one panel per stage.

Drawing *independent* samples from μ is the central computational problem: MD and MCMC mix slowly across the high torsion barriers that separate conformers, so a trained flow that emits independent conformers with a tractable density, and therefore supports the exact importance-sampling pipeline of the previous sections, is the payoff that justifies the training cost.

We take the potential energy to be the Amber molecular-mechanics force field (Cornell et al., 1995), the standard classical model, with raw force-field energy $U_{\text{FF}}(x)$ in kJ/mol, a sum of bonded terms, harmonic bonds and angles and periodic torsions, and nonbonded Lennard-Jones and Coulomb terms,

$$U_{\text{FF}}(x) = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 + \sum_{\text{torsions}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[4\epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^6}{r_{ij}^6} \right) + \frac{q_i q_j}{r_{ij}} \right], \quad (19)$$

the last sum running over nonbonded atom pairs, with charges in units that absorb the Coulomb constant. The Boltzmann distribution at temperature T is $\mu \propto e^{-U}$ in the dimensionless reduced potential

$$U(x) = \frac{1}{k_B T} U_{\text{FF}}(x),$$

with k_B the Boltzmann constant. We evaluate U_{FF} for the *isolated molecule in vacuum*, with no solvent and no nonbonded cutoff; this and two further simplifications of the molecular study are stressed in Section 6.5.

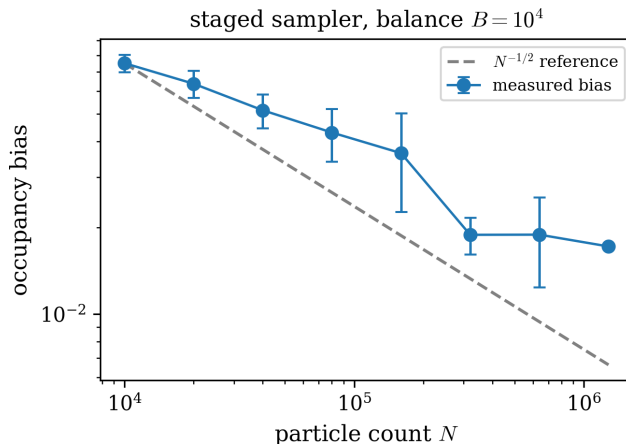


Figure 8: Occupancy bias of fresh staged rebuilds through the trained $B = 10^4$ ladder of the X-regularized loss, versus rebuild size $N = 10^4 \cdot 2^k$, $k = 0, \dots, 7$, averaged over 2^{7-k} independent rebuilds per size at equal total work; error bars are ± 2 standard errors. The dashed line is the Monte Carlo reference $N^{-1/2}$.

We work in *whitened internal coordinates* (Noé et al., 2019). The six rigid-body degrees of freedom, global translation and rotation, leave U unchanged, so we drop them and describe the shape by bond lengths, bond angles, and torsions, a z-matrix giving a smooth, invertible map $x \mapsto \xi \in \mathbb{R}^d$ with $d = 3N - 6$ for an N -atom molecule and a tractable volume Jacobian. Bonds and angles are stiff and near-Gaussian about their equilibria, so each is standardized to unit variance; torsions stay periodic on $[-\pi, \pi)$. The source $\mu_0 \propto e^{-U_0}$ is then a trivial product base, standard normal on the bond/angle directions and uniform on the torsions, and the target, pulled back through the change of variables, is

$$\mu(\xi) \propto \exp(-U(x(\xi))) \left| \det \left(\partial x / \partial \xi \right) \right|,$$

so U_0 and U as functions of ξ already absorb the change-of-variables log-Jacobian. This is what makes the target tractable for a flow: the rigid-body symmetries and the stiff, strongly correlated bond/angle vibrations are factored out, leaving the combinatorially multimodal torsional landscape, the conformers, as the structure the diffeomorphism must actually transport.

Two features make (19) qualitatively harder than a smooth model target, and they compound. First, the dimension grows with the molecule ($d = 36$ to 60 here), and the conformer landscape is *combinatorially multimodal*: every rotatable torsion is multi-well (gauche $^\pm$ /trans), so the joint over several torsions has dozens of basins the flow must discover. Second, and more consequential, the Lennard-Jones r^{-12} repulsion makes U *singular*: as any pair distance $r_{ij} \rightarrow 0$ the energy diverges, so the target has a hard, near-singular clash boundary that a smooth diffeomorphism cannot represent without enormous Jacobians. Neither difficulty is decisive in isolation, since high-dimensional smooth targets and low-dimensional singular ones are both routine, but together they are worse than either alone: the singular clash core sits *inside* the high-dimensional multimodal landscape, and a single flow asked to discover modes *and* resolve the singularity at once trains badly at both. This compounding of the two difficulties, absent from every model problem of the previous section, is what the rest of this section addresses. On alanine dipeptide ($d = 60$) below it is decisive: the clash core, not the torsion profile, becomes the binding constraint.

6.2 Sharpening: the core simplification

Sharpening keeps the singular part out of the flow entirely, by annealing a second index the flow never sees. Annealing runs from the source U_0 of Section 6.1 to the target, for which we now write the alias $U_1 = U$. A *regularization* index $s \in [0, 1]$ defines a soft-core target $U_{1,s}$ by taming the $r \rightarrow 0$ clash of the raw energy U_{FF} (19) before its reduction by $k_B T$. Two crucial soft-core parameters act on U_{FF} : an energy cap e_{cap} (kJ/mol) that log-compresses the energy above its threshold to $e_{\text{cap}} + a \log(1 + (U_{\text{FF}} - e_{\text{cap}})/a)$ with fixed scale $a = 50$ kJ/mol, a C^1 wall of finite height and gradient (a hard clamp would zero the escape force); and

a distance floor r_{floor} (nm) that replaces each nonbonded pair distance by $\max(r_{ij}, r_{\text{floor}})$, removing the r_{ij}^{-12} pole. Writing $U_{\text{FF}}|_{e_{\text{cap}}, r_{\text{floor}}}$ for the resulting regularized energy, the reduced soft-core target is

$$U_{1,s} = \frac{1}{k_B T} U_{\text{FF}}|_{e_{\text{cap}}(s), r_{\text{floor}}(s)}, \quad (20)$$

with the cap threshold $e_{\text{cap}}(s)$ and floor $r_{\text{floor}}(s)$ annealed from soft ($s = 0$) to sharp ($s = 1$) on a geometric or arithmetic schedule,

$$e_{\text{cap}}(s) = e_{\text{min}}(e_{\text{max}}/e_{\text{min}})^s \quad \text{or} \quad e_{\text{min}} + (e_{\text{max}} - e_{\text{min}})s, \quad r_{\text{floor}}(s) = r_{\text{max}} + (r_{\text{min}} - r_{\text{max}})s,$$

between a low cap e_{min} and high cap e_{max} , and a loose floor r_{max} and the physical floor r_{min} . Both controls act only on the high-energy clash region, inactive on the equilibrium basins where $U_{\text{FF}} \leq e_{\text{cap}}$ and $r_{ij} > r_{\text{floor}}$, so $U_{1,1} = U_1$ there and $U_{0,s} = U_0$ for all s . The bridge potential carries both indices,

$$U_{t,s} = (1-t)U_0 + tU_{1,s},$$

and the whole run is a single anneal from the soft source $U_{0,0}$ to the sharp target $U_{1,1}$, the regularization reaching the temperature at every accepted stage, $s_k = t_k$. The two indices coincide only at the accepted nodes: within a stage they differ, t first advancing to t_k while s lags at s_{k-1} , then s catching up to s_k , as described next.

The change from Algorithm 4 is one extra, flow-free step per stage. Algorithm 4 advances the single temperature index t on the bridge U_{t_k} ; with sharpening each stage advances the pair (t, s) in two sub-steps,

$$\underbrace{(t_{k-1}, s_{k-1}) \xrightarrow{G_k} (t_k, s_{k-1})}_{\text{flow step: advance } t} \xrightarrow{w_k^{\text{sharp}}} \underbrace{(t_k, s_{k-1}) \rightarrow (t_k, s_k)}_{\text{sharpening step: advance } s}.$$

The *flow step* advances the temperature at the *lagged* regularization, $U_{t_{k-1}, s_{k-1}} \rightarrow U_{t_k, s_{k-1}}$: at fixed s_{k-1} the soft-core bridge $U_{t, s_{k-1}}$ is a single-temperature bridge to the target $U_{1, s_{k-1}}$, so the trained map G_k and its validation update are exactly steps (i)–(v) of Algorithm 4 with U_1 replaced by $U_{1, s_{k-1}}$; the diffeomorphism only ever transports the smooth surrogate. The *Monte Carlo sharpening step* then advances the regularization at fixed temperature, $U_{t_k, s_{k-1}} \rightarrow U_{t_k, s_k}$, by a single importance reweight with weight

$$w_k^{\text{sharp}}(\xi) = \exp(U_{t_k, s_{k-1}}(\xi) - U_{t_k, s_k}(\xi)) = \exp(-t_k[U_{1, s_k}(\xi) - U_{1, s_{k-1}}(\xi)]), \quad (21)$$

followed by resampling and a short Langevin rejuvenation on U_{t_k, s_k} , with no flow. The importance step asks nothing the singularity forbids: it never inverts or differentiates U_1 , only reweights and rejuvenates samples inside the basins they already occupy, so the stiff repulsive wall costs a little weight variance rather than the unbounded Jacobian it would impose on a smooth map. This is the asymmetry the split exploits: the flow handles the part that is smooth but high-dimensional, the Monte Carlo step the singular part, confined to the thin clash boundary.

Each stage thus yields two effective sample sizes: the flow’s validation $\text{ESS}_k^{\text{val}}$ from the flow step, which is the ESS of the step-(v) validation weights w_k of Algorithm 4, and the sharpening $\text{ESS}_k^{\text{sharp}}$ of w_k^{sharp} (21); each is the normalized effective sample size of Section 3.4, a fraction in $(0, 1]$. The Monte Carlo error-propagation factor through the two per-stage reweights,

$$F = \prod_k \frac{1}{\text{ESS}_k^{\text{val}}} \frac{1}{\text{ESS}_k^{\text{sharp}}}, \quad F = 1 \text{ ideal}, \quad (22)$$

charges the cost of both; it is smaller-is-better, and a run with no sharpening step has $\text{ESS}_k^{\text{sharp}} \equiv 1$. On the clash-dominated target a few near-singular weights collapse the raw validation ESS, so before forming it we saturate the log-weights about their median. With m the median of $\{\log w_i\}$ and $\sigma_w = \text{cmedian}_i |\log w_i - m|$, the *smooth* validation effective sample size is

$$\text{ESS}_k^{\text{val}*} = \text{ESS}(\{\tilde{w}_i\}), \quad \log \tilde{w}_i = m + \sigma_w \tanh((\log w_i - m)/\sigma_w), \quad (23)$$

the normalized ESS of Section 3.4 evaluated on the saturated weights, with $c = 6$ on alanine dipeptide as detailed in Section 6.3. Its error-propagation factor is

$$F^* = \prod_k \frac{1}{\text{ESS}_k^{\text{val}*}} \frac{1}{\text{ESS}_k^{\text{sharp}}}. \quad (24)$$

We report F where the raw validation ESS is well behaved, and F^* , written with a star, where the clash singularity forces the smooth estimator; the two are the same product evaluated with raw or smooth validation ESS. This factor is the headline metric for every result below: the variance amplification a downstream importance-sampling estimator inherits from the full ladder.

6.3 Engineering for the singularity

The singular targets require three further devices, all inert on the smooth ones (glycerol and diethanolamine ran with none of them). (i) *Particle drop*: each validation and resample ESS discards the worst 2% of particles before the ESS is formed, so a handful of clash-singular outliers, finite but enormous under the soft cap, cannot hijack the importance weights. (ii) *Best ESS gate snapshots*: within a stage the flow is snapshotted at each ESS gate and the stage keeps the *highest* validation ESS snapshot rather than the last training step (which overfits the soft cap and sheds ESS); a step-0 identity snapshot is always included as a Monte Carlo fallback. (iii) *Distance-floor anneal*: for the worst clash core (alanine dipeptide) r_{floor} is annealed alongside the cap, softening the Lennard-Jones wall early and tightening it to the physical value only at $s = 1$, at the cost of a longer ladder. The wide-coverage pool $\hat{\mu}$ is supplied by the δ -reweighted quench and temper of Section 3.3 ($\delta = 0.1$), unchanged; the sharpening split remains the core mechanism, and these three devices are the supporting engineering the singularity demands.

A final, metric-level choice accompanies them on the singular target: the per-stage validation ESS that gates and scores each stage is the smooth estimator $\text{ESS}_k^{\text{val}*}$ of (23), with $c = 6$ on alanine dipeptide. Saturating the log-weights about their median bounds the few clash-singular dominators that otherwise collapse the raw weighted ESS; a raw ESS of 0.4 then reads as a smooth ≈ 0.5 . Glycerol and diethanolamine, free of such dominators, keep the raw ESS and report F .

6.4 Numerical results

We close three molecules of increasing difficulty in full internal coordinates on a single consumer GPU (RTX 5070 Ti): glycerol ($d = 36$), diethanolamine ($d = 48$), and alanine dipeptide ($d = 60$, the canonical Boltzmann generator benchmark). Table 7 summarizes the comparison that runs through all three: the X-regularized δ -reweighted loss against an annealed SMC reference, the same adaptive sharpening scheme with the flow set to the identity at every stage, which selects its own, longer ladder. Because the factor is a product over stages, the *ratio* of trained-flow to annealed SMC F , not the absolute value, is the cross-molecule comparison: the trained flow cuts F by $\sim 30\text{--}40\times$ on the two clash-mild, combinatorially multimodal targets, but only $\sim 2\times$ on alanine dipeptide, where the singular clash core rather than the multimodality dominates — the core that the Monte Carlo sharpening already resolves. The per-molecule subsections below give the full ablation, ladder, and reference figures for each. The baselines here, the bare forward KL and the identity-flow annealed SMC, are the natural same-cost comparisons for a forward KL-trained generator.

6.4.1 Glycerol ($d = 36$)

We ablate the generator on the full internal-coordinate glycerol target along three independent axes: the loss (forward KL versus the X-regularized $\text{KL} + \text{X}_{\mu} + \text{X}_{(\hat{\mu} + \bar{v})/2}$); the regularization schedule (*sharpen*: the soft-core cap e_{cap} is geometrically annealed from 100 to 200 kJ/mol along the ladder with a Monte Carlo sharpening step per stage; *raw*: fixed $e_{\text{cap}} = 200$ kJ/mol, no anneal or sharpening); and the wide-coverage sampler (plain quench and temper versus its δ -reweighted variant, $\delta = 0.1$, defined only for the X loss). The headline is the error-propagation factor F (22); the *raw* runs take no sharpening step ($\text{ESS}^{\text{sharp}} \equiv 1$).

The X regularizer is the dominant factor (Table 8), roughly halving F at either schedule. The δ -reweighted quench and temper lowers F further in both schedules. Once the sharpening reweight is charged to F , the *sharpen* schedule no longer beats the fixed-cap *raw* run on F alone; its advantage is feasibility and speed.

target	d	K	F/F^*	
			KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ flow	annealed SMC
glycerol	36	4	14.16	463.02
diethanolamine	48	5	35.90	1455.44
alanine dipeptide	60	11	67.95*	129.70*

Table 7: Molecular Boltzmann generators in full internal coordinates. K is the trained-flow ladder length; the annealed SMC reference adaptively chooses its own, longer ladder, $K = 10, 10, 12$ respectively. Both columns run the same adaptive sharpening scheme, annealing the regularization from the soft source to the sharp target, and differ in whether each stage trains a flow or sets it to the identity. Entries are the error-propagation factor F (22), or the smooth ESS variant F^* (24) where starred, used on alanine dipeptide; both include the Monte Carlo sharpening reweight, smaller is better, best in bold.

loss (raw pushforward)	K	wall (min)	F
annealed SMC	10	2	635.56
forward KL	7	215	40.86
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	6	162	21.96
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	6	158	16.56
loss (sharpening)	K	wall (min)	F
annealed SMC	10	5	463.02
forward KL	4	88	44.00
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	4	73	22.51
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	4	73	14.16

Table 8: Glycerol ($d = 36$) ablation, raw pushforward (top) and sharpening (bottom): ladder length K , training wall-clock, and the error-propagation factor F (22) (smaller better; the sharpening rows include the per-stage Monte Carlo sharpening reweight). δ -rw. = δ -reweighted quench and temper.

The lowest- F and fastest run is KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$, δ -reweighted, under the sharpen schedule. Turning the trained flow off entirely (the identity map at every stage, i.e. plain adaptive annealed SMC over the same ladder and acceptance standard) gives a far larger F in $K = 10$ stages. The trained flow thus cuts the propagation factor by ~ 30 – $40\times$ while shortening the ladder from $K = 10$ to $K = 4$ – 6 (Figure 9): this is the quantitative value the transport map adds over annealing alone.

Glycerol is achiral (each torsion marginal is even, $p(\phi) = p(-\phi)$; Figure 10), and every rotatable torsion is three-fold: gauche $^{\pm}$ near $\pm 60^\circ$ and trans near 180° . The $\pm\phi$ symmetry pairs gauche $^- \leftrightarrow$ gauche $^+$ and leaves trans self-symmetric, so the joint target over the five rotatable torsions is combinatorially multimodal (dozens of conformers), which is what the wide-coverage pool must cover.

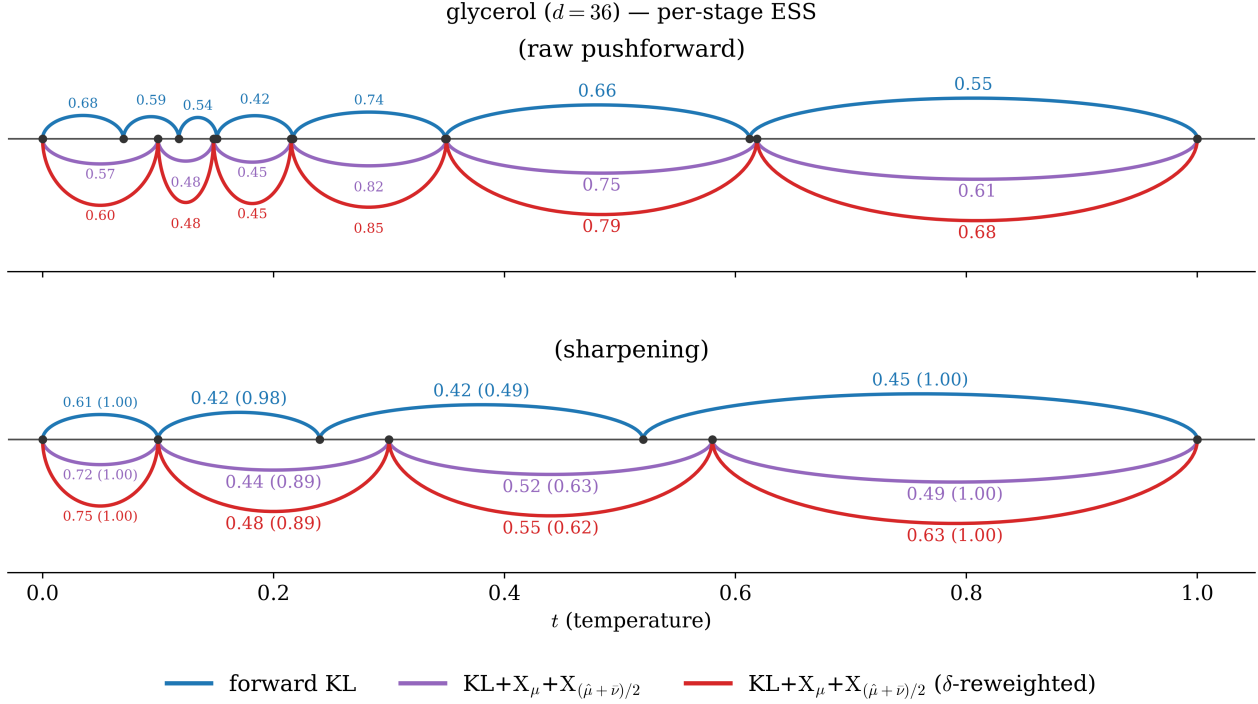


Figure 9: Glycerol per-stage validation ESS as annealing-ladder arcs, raw (top) and sharpen (bottom). In each row forward KL is drawn above the axis (blue), $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ below (purple), and its δ -reweighted variant further below (red). Each arc is an accepted stage; the apex label is that stage’s validation ESS; a cross marks a bridge the ladder failed to reach.

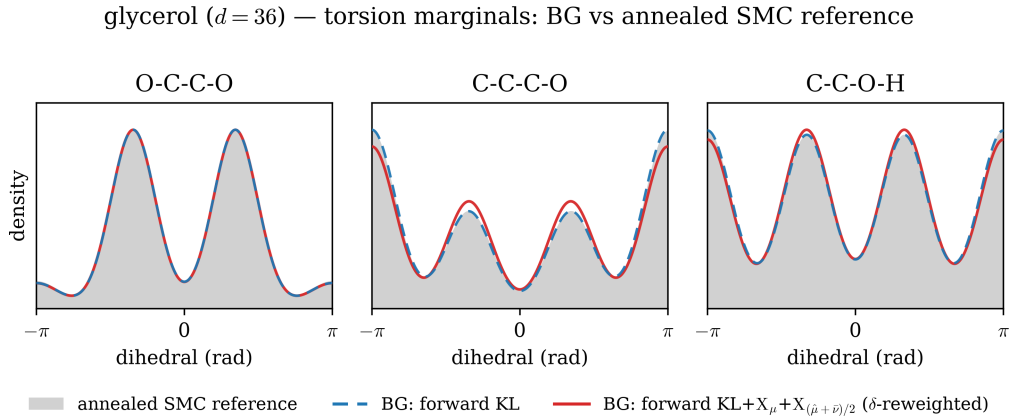


Figure 10: Glycerol torsion marginals of both generators against a converged annealed SMC reference (grey): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted (red), both sharpened with the cap annealed to $e_{\text{cap}} = 200$ kJ/mol. Both recover the multimodal gauche $^\pm$ /trans structure of the characteristic glycerol torsions.

6.4.2 Diethanolamine ($d = 48$)

We test the generator on the full internal-coordinate diethanolamine target ($\text{HN}(\text{CH}_2\text{CH}_2\text{OH})_2$, $d = 48$), comparing the two best losses under the same sharpening schedule (the soft cap geometrically annealed $e_{\text{cap}} : 100 \rightarrow 200$ kJ/mol, a Monte Carlo sharpening step per stage): forward KL, and the X-regularized $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ with the δ -reweighted quench and temper ($\delta = 0.1$). Forward KL reaches $t = 1$ in $K = 6$ stages, the X-regularized loss in $K = 5$, a single direct $0.65 \rightarrow 1$ step.

loss	K	wall (min)	F
annealed SMC	10	7	1455.44
forward KL	6	896	92.60
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	5	782	35.90

Table 9: Diethanolamine ($d = 48$): two trained-flow losses and an annealed SMC reference, all with a per-stage Monte Carlo sharpening reweight (cap e_{cap} geometrically annealed $100 \rightarrow 200$ kJ/mol): ladder length K , training wall-clock, and the error-propagation factor F (22) (smaller better; F includes the sharpening reweight). δ -rw. = δ -reweighted quench and temper.

The X-regularized loss with δ -reweighting reaches a lower F than forward KL (Table 9), a $\sim 2.6\times$ reduction in importance-weight waste at identical settings (glycerol showed a comparable $\sim 3.1\times$ gap). The advantage shows in the late, sharp stages and in the ladder length: forward KL’s per-stage validation ESS stalls and it needs 6 rungs to reach $t = 1$, whereas the $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted run holds a higher ESS and reaches $t = 1$ in 5 (Figure 11). This is a singular, high-dimensional, combinatorially multimodal target, so each run is a multi-hour computation; closing it efficiently is the practical contribution of the X-regularized loss.

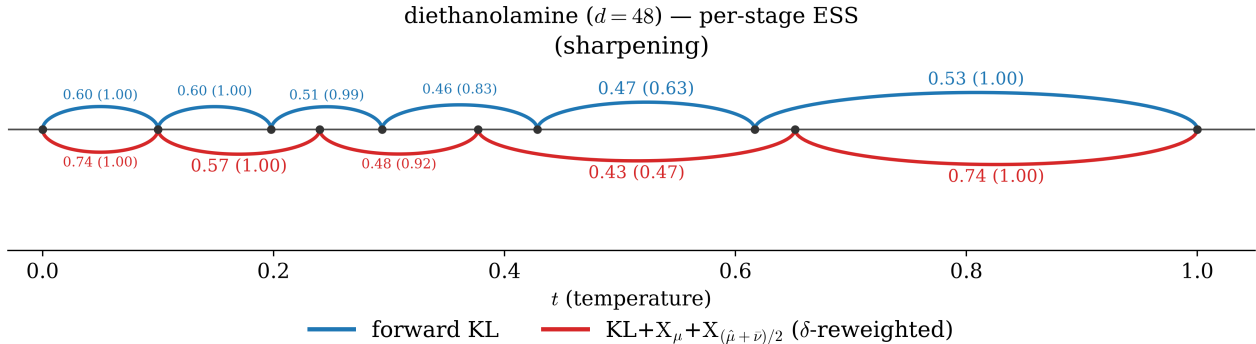


Figure 11: Diethanolamine per-stage ESS as annealing-ladder arcs (sharpening schedule): forward KL above the axis (blue), $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted, below (red). Each arc is an accepted stage; the apex label is that stage’s validation ESS, with the sharpening ESS in parentheses.

Diethanolamine has a plane of symmetry (each torsion marginal is even, $p(\phi) = p(-\phi)$); its rotatable backbone torsions, O-C-C-N and C-C-N-C along the ethanolamine arms plus the hydroxyl rotations, are each multi-well (gauche $^\pm$ /trans), so the joint target is combinatorially multimodal, which the wide-coverage pool must cover.

Against the annealed SMC reference (Figure 12) the two losses are visually indistinguishable: both recover the multimodal backbone torsions and neither is discernibly closer to the reference. The separation is in efficiency, not in the sampled marginals: $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ reaches $t = 1$ in fewer stages with a higher final-step ESS, a $\sim 2.6\times$ smaller error-propagation factor, and fewer rejected stage attempts in the early ladder. Switching the trained flow off entirely (the identity map at every stage, i.e. plain adaptive annealed SMC over the same ladder) gives a far larger F . The trained X-regularized flow therefore cuts the propagation factor by $\sim 40\times$ and halves the ladder length (5 vs 10): on this singular target the transport map adds even more over annealing alone than on glycerol, a justification for training a flow.

diethanolamine ($d = 48$) — torsion marginals: BG vs annealed SMC reference

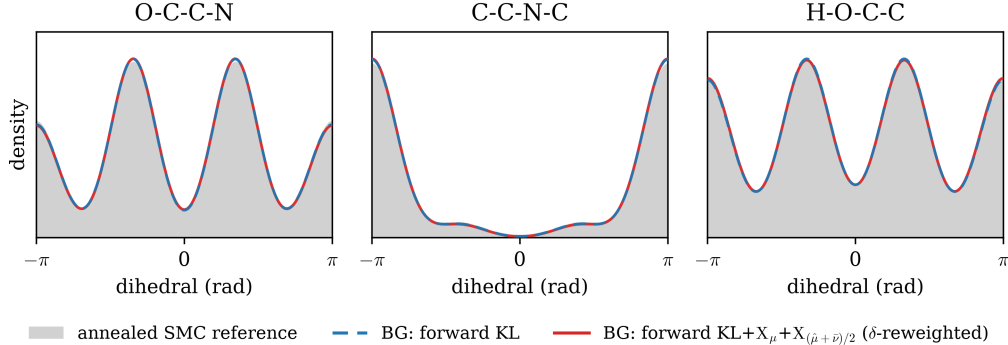


Figure 12: Diethanolamine torsion marginals of both generators against a converged annealed SMC reference (grey): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted (red), both sharpened with the cap annealed to $e_{\text{cap}} = 200$ kJ/mol. Both recover the multimodal backbone torsions and track the annealed SMC reference.

6.4.3 Alanine dipeptide ($d = 60$)

We push the generator to alanine dipeptide (Ac-Ala-NHMe), the canonical Boltzmann generator benchmark, in full internal coordinates ($d = 60$). The qualitative step up from glycerol and diethanolamine is the target’s character: those are dominated by smooth rotatable-bond multimodality, whereas alanine dipeptide folds the backbone tightly enough that the Lennard-Jones clash core, not the torsion profile, sets the hardest part of the anneal. We report the X-regularized loss only ($\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted, $\delta = 0.1$), with a per-stage Monte Carlo sharpening in which *both* the cap $e_{\text{cap}} : 100 \rightarrow 200$ kJ/mol and the clash floor $r_{\text{floor}} : 0.2 \rightarrow 0.1$ nm are annealed arithmetically, unlike the geometric e_{cap} schedule of glycerol and diethanolamine; alanine dipeptide is also the only target that anneals r_{floor} , which glycerol and diethanolamine hold fixed at 0.1 nm. The run reaches $t = 1$ in $K = 11$ stages, with every stage holding a high validation and sharpening ESS (Table 10, Figure 13). The bare composed flow’s end-to-end ESS against the sharp target, from a single importance reweight of source samples pushed through the fully composed K -stage map and scored against $U_{1,1}$, is only $\sim 5 \times 10^{-6}$, so the per-stage sharpening, not a raw pushforward, is the deliverable.

loss	K	wall (min)	F^*
annealed SMC	12	23	129.70*
$\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	11	~ 1000	67.95*

Table 10: Alanine dipeptide ($d = 60$), the X-regularized loss with a per-stage Monte Carlo sharpening reweight (the cap e_{cap} and floor r_{floor} annealed arithmetically): ladder length K , wall-clock, and the error-propagation factor F^* (24), starred throughout because every entry uses the smooth validation ESS (smaller better; F^* includes the sharpening reweight). The first row is an annealed SMC reference, the identity map over the same ladder. δ -rw. = δ -reweighted quench and temper.

For comparison with the lighter targets, alanine dipeptide’s larger F^* is the price of its singular nonbonded core, not of its multimodality. Three alanine-dipeptide-only factors nonetheless make this F^* read smaller than a strict like-for-like extension of that trend: the per-stage ESS is the smooth estimator $\text{ESS}_k^{\text{val}*}$ (23) rather than the raw weighted ESS; r_{floor} is annealed alongside the cap; and the ladder is much longer ($K = 11$ versus 4 and 5), its smaller per-step jumps keeping each per-stage ESS high. All three raise the per-stage ESS entering F^* , and none applied to the lighter targets.

The instability is sharply localized to a clash bottleneck near $t \approx 0.3$ – 0.43 . The stages spanning $t \approx 0.29$ – 0.43 (soft cap $e_{\text{cap}} \approx 129$ – 143 kJ/mol) are where the nonbonded core first bites: the run minimum validation ESS sits here, and these are the *only* stages that shrink the step or retry heavily. Below $t \approx 0.3$ the torsions still dominate and the flow trains in a single attempt; above $t \approx 0.5$ the cap is sharp enough that the basins separate cleanly and ESS recovers to ≈ 0.8 . The binding constraint for alanine dipeptide is the clash

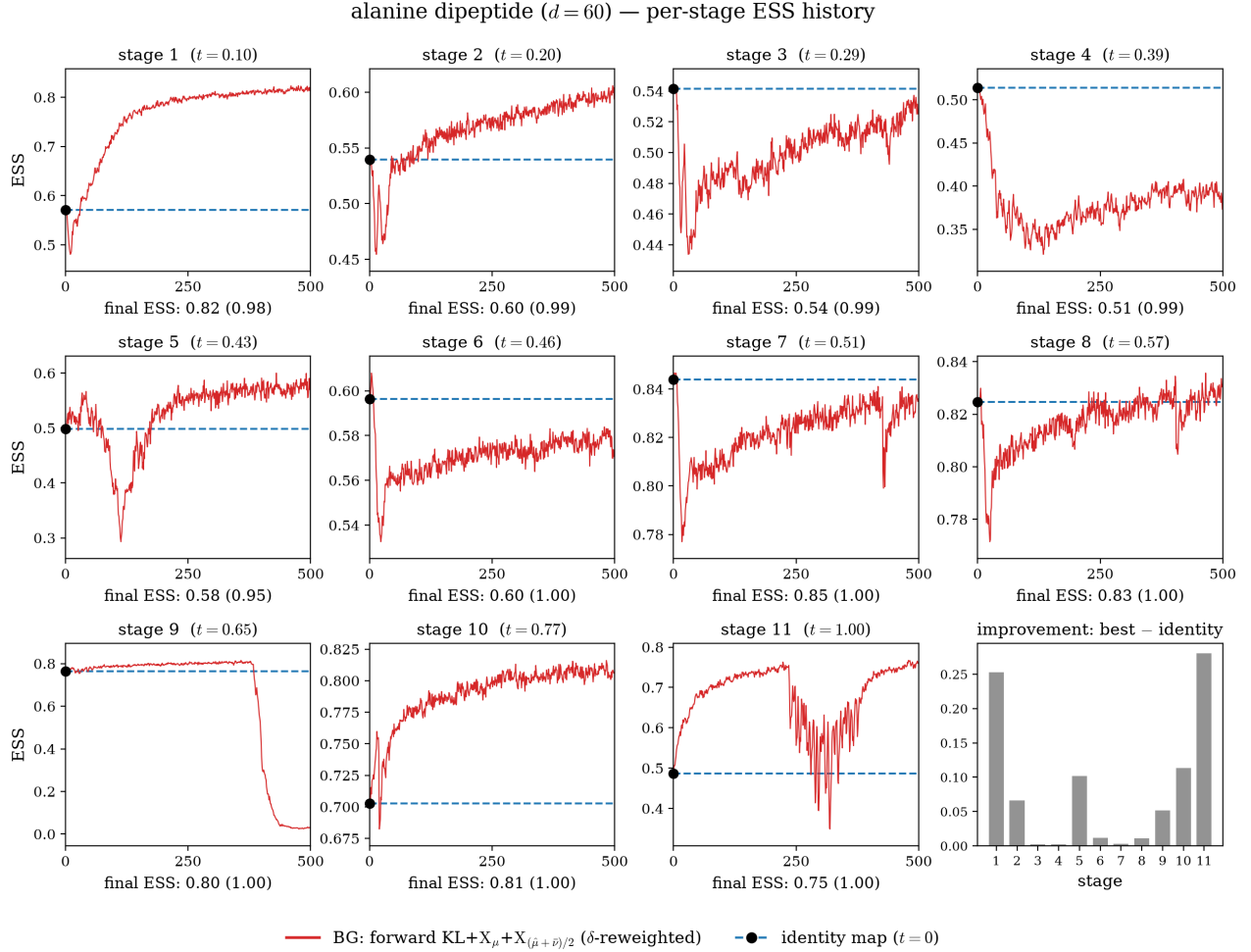


Figure 13: Alanine dipeptide per-stage ESS history: each panel is the per-step effective sample size during that stage’s X-regularized δ -reweighted flow training (red), against the identity-map baseline at $t = 0$ (blue dashed). Below each panel is that stage’s accepted validation ESS with the sharpening ESS in parentheses. The dip across $t \approx 0.29$ – 0.43 is the clash bottleneck, with ESS recovering to ≈ 0.8 once the cap is sharp enough to separate the basins.

singularity, not the multimodality: the qualitative difference from the $d \leq 48$ molecules, and the reason the three devices of Section 6.3 (particle drop, best ESS gate snapshots, r -floor anneal), inert on glycerol and diethanolamine, were necessary here.

The per-stage ESS history (Figure 13) exposes two coupled instabilities that the best ESS gate works around but does not cure. First, the within-stage ESS is *non-monotone and unstable*: it can rise, plateau, then drop abruptly (the sharp collapse late in the $t = 0.65$ stage and the volatility at $t = 1$ are the clearest cases), so the last training step is frequently far below the best. Second, near the clash bottleneck ($t \approx 0.3$ – 0.5) several flows *degenerate toward the identity map*: the best achievable ESS barely exceeds the $t = 0$ identity baseline, i.e. training fails to improve on doing nothing. Both are understood consequences of the *singularity of the potential*: the Lennard-Jones $r \rightarrow 0$ divergence leaves the soft-capped target ill-conditioned exactly where the nonbonded core first bites, so the gradient signal that would pull the flow off identity is weakest there. We report this as a known limitation; robust remedies (stronger or adaptive regularization annealing, parameterizations tailored to the repulsive wall) are left to future work.

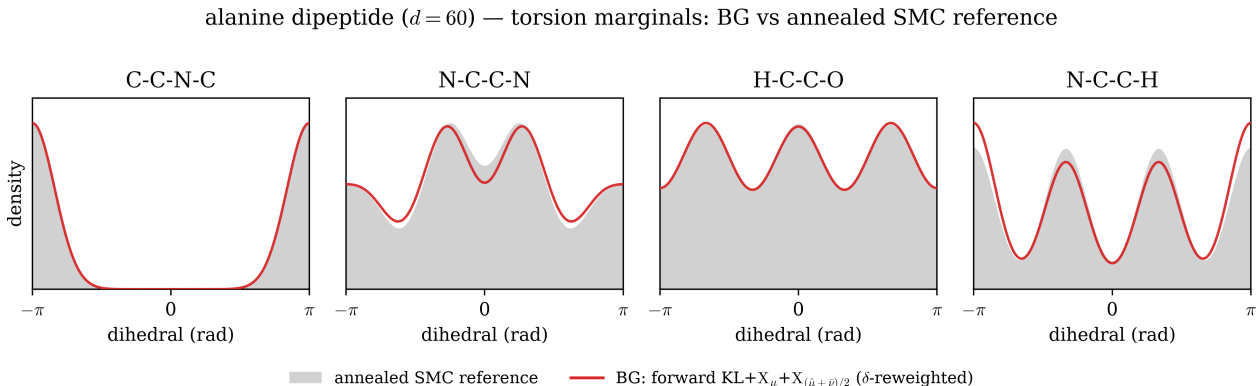


Figure 14: Alanine dipeptide torsion marginals of the BG deliverable (red) against a converged annealed SMC reference (grey). The four most multimodal proper torsions are shown; the densities are $\pm\phi$ -symmetrised, the achiral target having $p(\phi) = p(-\phi)$. The BG recovers the multimodal backbone torsions and tracks the reference; the peptide-bond torsion ω (C-C-N-C) is trans-dominant in the generator, the one torsion whose high rotation barrier also makes the uniform-init annealed SMC reference slow to converge.

alanine dipeptide’s backbone ϕ, ψ are multi-basin (Figure 14), and a second, alanine-dipeptide-specific feature is chirality: the classical force field is mirror-symmetric (every term in (19) is invariant under reflection), so the L- and D-forms are isoenergetic, and the internal-coordinate flow, carrying no chirality constraint, emits *both* where a physical trajectory would stay in one.

The X-regularized BG closes the full internal-coordinate alanine dipeptide target to $t = 1$ in 11 stages, recovering the multimodal backbone torsions against a converged annealed SMC reference and emitting both chiral forms of the mirror-symmetric target. The lesson carried over from the lighter $d \leq 48$ molecules is inverted: alanine dipeptide’s difficulty is not its combinatorial torsion multimodality but its nonbonded clash core, concentrated in the $t \approx 0.3$ – 0.43 stages, which the particle drop, best ESS gate snapshots, and r -floor anneal together make tractable. Across all three molecules the pattern is consistent: the sharpening split makes the singular target trainable at all, and the X-regularized flow then earns a ~ 30 – $40\times$ reduction in importance-weight waste where the landscape is multimodal but clash-mild, narrowing to $\sim 2\times$ where the singularity itself dominates.

6.5 Limitations

Four deliberate simplifications bound the molecular experiments, each a direction for future work. First, the targets are evaluated in vacuum: U_{FF} is the isolated-molecule Amber potential (19) with no solvent and no nonbonded cutoff. Solvation only reshapes U_{FF} and not the sampler, and the vacuum targets already display the well-separated potential wells the generator must discover and resolve, but implicit or explicit solvent and larger peptides are left untreated. Second, the internal-coordinate generator carries no chirality constraint, so for the achiral Amber force field it samples both the L- and D-enantiomers (Section 6.4.3), whereas a

physical trajectory stays in one; isolating a single enantiomer by a filter or restraint is left to future work. Third, we do not benchmark against FAB (Midgley et al.): its large per-step batches and replay buffer make a like-for-like comparison under the same compute budget as these forward KL-type losses prohibitive, so the natural same-cost baselines here are the bare forward KL and the identity-flow annealed SMC. Fourth, on the alanine dipeptide clash core the per-stage flow training is unstable and can degenerate toward the identity map, as documented in Section 6.4.3; the best ESS gate works around this but does not cure it, and a regularization or parameterization that conditions the repulsive wall remains open.

7 Conclusions

Forward KL training of a normalizing flow Boltzmann generator usually trains well, but it can silently miss modes and report a fake effective sample size, and noisy or biased annealed samples limit its accuracy; a smaller leakage mechanism, where the target vanishes, rides alongside. We close these gaps by regularizing the forward KL with a term that reuses the flow’s own log-ratio.

- We propose the X functionals. They augment the forward KL by penalizing the pairwise variation of $\log(\mu/\nu)$ under a chosen non-negative weight. One forward pass of the flow computes them; the shape term costs almost nothing beyond a plain forward KL step, while the mixture adds a further sampling batch.
- One regularizer takes three complementary roles through its weight. The weight X_μ controls the shape of the log-ratio and stabilizes training. The weight $X_{\hat{\mu}}$ drives mode discovery, drawing the flow toward a wide-coverage measure that quench and temper prepares before training. The weight $X_{\hat{\nu}}$ suppresses leakage.
- The balanced loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$ combines the three roles in one objective, folding mode discovery and leakage suppression into a single pairwise integral. It serves as a sensible default and needs little tuning. A Fisher–Rao analysis shows the forward KL still decays along the X_μ -regularized flow, and under a bound on the regularization strength the X_μ term can lower the accuracy floor that biased AIS samples leave behind. The forward KL is sensitive to these samples early in training, so the X-regularized losses are particularly useful when the training batch is small. The loss applies across smooth benchmarks, a high-dimensional multi-well, and lattice field theories, where it tends to discover modes and sharpen accuracy.
- We build an adaptive-temperature Boltzmann generator on the loss. It grows its ladder from the SMC effective sample size; on molecular potentials, a sharpening reweight pulls each stage toward the physical potential. In our experiments it tends to improve per-stage quality and reduce importance-weight waste over annealing alone, with smaller gains where a singular clash core, not multimodality, is the difficulty.

Several directions invite further study. First, a careful comparison with FAB (Midgley et al.), and a possible combination with it, would weigh our quench against its replay buffer, and the same comparison extends to other AIS-based flow samplers. Second, the training instability at the singular clash core calls for a remedy that conditions the repulsive wall, since the present best ESS gate works around the instability rather than curing it. Third, larger and solvated systems are a natural target, where the potential reshapes but the sampler need not change. Finally, a JAX implementation in place of the present PyTorch one would speed up both the training and the inference.

Acknowledgement

The Python codes implementing the normalizing flows are built upon the Python packages `zflows` and `zflows_md`, written by the author Xuda Ye and installable through Anaconda; the source code reproducing all numerical experiments is openly available. Both are listed below:

```
conda install xudaye::zflows xudaye::zflows.md  
https://github.com/xuda-ye-math/X-regularization
```

Numerical tests are implemented on an NVIDIA RTX 5070 Ti GPU with 16 GB of VRAM.

References

- Tara Akhound-Sadegh, Jarrod Rector-Brooks, Avishek Joey Bose, Sarthak Mittal, Pablo Lemos, Cheng-Hao Liu, Marcin Sendera, Siamak Ravanbakhsh, Gauthier Gidel, Yoshua Bengio, Nikolay Malkin, and Alexander Tong. Iterated denoising energy matching for sampling from Boltzmann densities. In *International Conference on Machine Learning*, 2024.
- Tara Akhound-Sadegh, Jungyoon Lee, Avishek Joey Bose, Valentin De Bortoli, Arnaud Doucet, Michael Bronstein, Dominique Beaini, Siamak Ravanbakhsh, Kirill Neklyudov, and Alexander Tong. Progressive inference-time annealing of diffusion models for sampling from Boltzmann densities. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Michael Samuel Albergo and Eric Vanden-Eijnden. NETS: A non-equilibrium transport sampler. In *International Conference on Machine Learning*, pages 1026–1055. PMLR, 2025. arXiv preprint arXiv:2410.02711.
- Michael Arbel, Alexander G. D. G. Matthews, and Arnaud Doucet. Annealed flow transport Monte Carlo. In *International Conference on Machine Learning (ICML)*, 2021.
- Julius Berner, Lorenz Richter, and Karen Ulrich. An optimal control perspective on diffusion-based generative modeling. *Transactions on Machine Learning Research*, 2024.
- Omar Chehab, Anna Korba, Austin Stromme, and Adrien Vacher. Provable convergence and limitations of geometric tempering for Langevin dynamics. In *International Conference on Learning Representations*, volume 2025, pages 5451–5493, 2025.
- Wendy D. Cornell, Piotr Cieplak, Christopher I. Bayly, Ian R. Gould, Kenneth M. Merz, David M. Ferguson, David C. Spellmeyer, Thomas Fox, James W. Caldwell, and Peter A. Kollman. A second generation force field for the simulation of proteins, nucleic acids, and organic molecules. *Journal of the American Chemical Society*, 117(19):5179–5197, 1995.
- Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006.
- Arnaud Doucet, Nando De Freitas, and Neil Gordon. An introduction to sequential Monte Carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.
- Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows. *Advances in neural information processing systems*, 32, 2019.
- Víctor Elvira, Luca Martino, and Christian P Robert. Rethinking the effective sample size. *International Statistical Review*, 90(3):525–550, 2022.
- Loris Felardos, Jérôme Hénin, and Guillaume Charpiat. Designing losses for data-free training of normalizing flows on Boltzmann distributions. *arXiv preprint arXiv:2301.05475*, 2023.
- Marylou Gabrié, Grant M Rotskoff, and Eric Vanden-Eijnden. Adaptive Monte Carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- Charles J. Geyer. Markov chain Monte Carlo maximum likelihood. In *Computing Science and Statistics: Proc. 23rd Symposium on the Interface*, pages 156–163, 1991.
- W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1):97–109, 1970.

- Aaron Havens, Benjamin Kurt Miller, Bing Yan, Carles Domingo-Enrich, Anuroop Sriram, Brandon Wood, Daniel Levine, Bin Hu, Brandon Amos, Brandon Karrer, Xiang Fu, Guan-Horng Liu, and Ricky T. Q. Chen. Adjoint sampling: highly scalable diffusion samplers via adjoint matching. In *International Conference on Machine Learning (ICML)*, 2025.
- Jiajun He, Yuanqi Du, Francisco Vargas, Dinghuai Zhang, Shreyas Padhy, RuiKang OuYang, Carla Gomes, and José Miguel Hernández-Lobato. No trick, no treat: pursuits and challenges towards simulation-free training of neural samplers. In *ICLR 2025 Workshop on Frontiers in Probabilistic Inference*, 2025.
- Koji Hukushima and Koji Nemoto. Exchange Monte Carlo method and application to spin glass simulations. *Journal of the Physical Society of Japan*, 65(6):1604–1608, 1996.
- Jorge V. José, Leo P. Kadanoff, Scott Kirkpatrick, and David R. Nelson. Renormalization, vortices, and symmetry-breaking perturbations in the two-dimensional planar model. *Physical Review B*, 16(3):1217–1241, 1977.
- Durk P Kingma and Prafulla Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems*, 31, 2018.
- Leon Klein and Frank Noé. Transferable Boltzmann generators. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Leon Klein, Andreas Krämer, and Frank Noé. Equivariant flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Jonas Köhler, Leon Klein, and Frank Noé. Equivariant flows: exact likelihood generative learning for symmetric densities. In *International Conference on Machine Learning (ICML)*, 2020.
- Jun S Liu. *Monte Carlo strategies in scientific computing*, volume 10. Springer, 2001.
- Alex Matthews, Michael Arbel, Danilo J. Rezende, and Arnaud Doucet. Continual repeated annealed flow transport Monte Carlo. In *ICML*, 2022.
- Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.
- Laurence Illing Midgley, Vincent Stimper, Gregor NC Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. In *NeurIPS 2022 AI for Science: Progress and Promises*.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- Christian A. Naesseth, Fredrik Lindsten, and David M. Blei. Markovian score climbing: variational inference with $KL(p||q)$. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Radford M. Neal. Annealed importance sampling. *Statistics and Computing*, 11(2):125–139, 2001.
- Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Nikolas Nüsken and Lorenz Richter. Solving high-dimensional Hamilton–Jacobi–Bellman PDEs using neural networks: perspectives from controlled diffusions and measures on path space. *Partial Differential Equations and Applications*, 2:48, 2021.
- George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. *Advances in neural information processing systems*, 30, 2017.

- Danyal Rehman, Oscar Davis, Jiarui Lu, Jian Tang, Michael Bronstein, Yoshua Bengio, Alexander Tong, and Avishek Joey Bose. Efficient regression-based training of normalizing flows for Boltzmann generators. In *International Conference on Learning Representations (ICLR)*, 2026.
- Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR, 2015.
- Lorenz Richter and Julius Berner. Improved sampling via learned diffusions. In *International Conference on Learning Representations (ICLR)*, 2024.
- Henrik Schopmans and Pascal Friederich. Temperature-annealed Boltzmann generators. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2501.19077.
- Henrik Schopmans, Annika von Klitzing, and Pascal Friederich. Efficient training of Boltzmann generators using off-policy log-dispersion regularization. *arXiv preprint arXiv:2602.03729*, 2026.
- John Skilling. Nested sampling for general Bayesian computation. *Bayesian Analysis*, 1(4):833–859, 2006.
- Nikola Surjanovic, Saifuddin Syed, Alexandre Bouchard-Côté, and Trevor Campbell. Parallel tempering with a variational reference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Robert H. Swendsen and Jian-Sheng Wang. Replica Monte Carlo simulation of spin-glasses. *Physical Review Letters*, 57(21):2607–2609, 1986.
- Saifuddin Syed, Alexandre Bouchard-Côté, George Deligiannidis, and Arnaud Doucet. Non-reversible parallel tempering: a scalable highly parallel MCMC scheme. *Journal of the Royal Statistical Society Series B*, 84(2):321–350, 2022.
- Charlie B. Tan, Avishek Joey Bose, Chen Lin, Leon Klein, Michael M. Bronstein, and Alexander Tong. Scalable equilibrium sampling with sequential Boltzmann generators. In *International Conference on Machine Learning (ICML)*, 2025.
- Lorenz Vaitl, Kim A. Nicoli, Shinichi Nakajima, and Pan Kessel. Gradients should stay on path: better estimators of the reverse- and forward-KL divergence for normalizing flows. *Machine Learning: Science and Technology*, 2022.
- Francisco Vargas, Will Grathwohl, and Arnaud Doucet. Denoising diffusion samplers. In *International Conference on Learning Representations*, 2023.
- Annika von Klitzing, Denis Blessing, Henrik Schopmans, Pascal Friederich, and Gerhard Neumann. Learning Boltzmann generators via constrained mass transport. In *International Conference on Learning Representations (ICLR)*, 2026.
- Martin J. Wainwright and Michael I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008.
- Peter Wirnsberger, Andrew J. Ballard, George Papamakarios, Stuart Abercrombie, Sébastien Racanière, Alexander Pritzel, Danilo Jimenez Rezende, and Charles Blundell. Targeted free energy estimation via learned mappings. *The Journal of Chemical Physics*, 153(14), 2020.
- Dongyeop Woo and Sungsoo Ahn. Iterated energy-based flow matching for sampling from Boltzmann densities. In *NeurIPS*, 2024. arXiv:2408.16249.
- Dawn B. Woodard, Scott C. Schmidler, and Mark Huber. Conditions for rapid mixing of parallel and simulated tempering on multimodal distributions. *The Annals of Applied Probability*, 19(2):617–640, 2009.
- Hao Wu, Jonas Köhler, and Frank Noé. Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.

Leo Zhang, Peter Potapchik, Jiajun He, Yuanqi Du, Arnaud Doucet, Francisco Vargas, Hai-Dang Dau, and Saifuddin Syed. Accelerated parallel tempering via neural transports. *arXiv preprint arXiv:2502.10328*, 2025.

Qinsheng Zhang and Yongxin Chen. Path integral sampler: A stochastic control approach for sampling. In *International Conference on Learning Representations*, 2022.